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For correspondence:

keiko.kono@oist.jp

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Pablo S Aguilar,
Instituto de Fisiología Biología
Molecular y Neurociencias (IFIBYNE),
Argentina

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Large-scale identification of plasma membrane repair proteins revealed spatiotemporal cellular responses to plasma membrane damage

Yuta Yamazaki, Keiko Kono 

Okinawa Institute of Science and Technology Graduate University, Onna, Japan

eLife Assessment

This work provides an **important** resource identifying 72 proteins as novel candidates for plasma membrane and/or cell wall damage repair in budding yeast, and describes the temporal coordination of exocytosis and endocytosis during the repair process. The data are **convincing**; however, additional experimental validation will better support the claim that repair proteins shuttle between the bud tip and the damage site.

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Abstract

Damage to the plasma membrane (PM) is common in all types of cells. PM repair processes, including exocytosis and endocytosis, are not mutually exclusive; rather, they collaborate to repair the wound. However, the temporal coordination between the repair processes remains poorly understood. Here, by large-scale identification and live-cell imaging of PM repair proteins, we analyzed the spatiotemporal PM damage responses in *Saccharomyces cerevisiae*. Of the 80 repair proteins identified, 72 proteins were previously unreported repair protein candidates. Among the observed repair processes, the polarized exocytosis and clathrin-mediated endocytosis (CME) are coupled at the damage site, with exocytosis predominating in the early stage of PM repair and CME predominating in the late stage of PM repair. Furthermore, we showed that CME at the growing bud site directs PM repair proteins with transmembrane domains to the damage site. We propose a model in which CME delivers repair proteins with transmembrane domains between the growing bud site and the damage site. This study provides a functional catalog of PM repair proteins and insights into spatiotemporal cellular responses to PM damage.

Introduction

The plasma membrane (PM) functions as a defensive barrier in all types of cells to protect cellular components from extracellular stimuli. PM frequently experiences physiological and pathological damages, such as eccentric contraction-induced injuries, migration-induced injuries, pore formation by bacterial toxins, or invasion by cancer cells (1, 2). Cells undergo cell lysis if the damaged PM is not repaired. Deficits in PM repair are linked to multiple diseases, including limb-girdle muscular dystrophy (3) and Scott syndrome (4, 5). Therefore, virtually all cells have PM repair machinery.

In eukaryotes, the influx of Ca^{2+} triggers PM repair (1). Upon Ca^{2+} influx, fundamental cellular mechanisms, including exocytosis (6–10), endocytosis (11, 12), membrane shedding by ESCRT complex (13, 14), and the constriction forces by actin cytoskeleton (15–18) are directed to the damage site to reseal the wound. These initial repair processes usually occur within 1 min after PM damage (19). After the resealing of the damaged PM, cells restructure the damaged PM (19–21). Restructuring is defined as the process by which cells modify the damaged PM to restore PM

homeostasis and normal cell function (19–21). These repair processes are not mutually exclusive, and they collaborate to repair the damaged PM (22). However, the temporal coordination between the PM repair processes remains poorly understood.

Both resealing and restructuring of the damaged PM are mediated by the proteins, which accumulate at the damage site. Here, we defined them as PM repair proteins. PM repair proteins include ESCRT, annexins, and dysferlin in mammalian cells (13, 23–25). Using budding yeast *Saccharomyces cerevisiae* as a model, we previously identified evolutionarily conserved repair proteins, including protein kinase C, exocyst, and phospholipid flippases (26, 27). These repair proteins accumulate at the damage site at the appropriate time and proceed with the repair processes. Therefore, identifying repair proteins and observing their movements after PM damage is an important step toward understanding the temporal PM repair processes.

Here, we performed proteome-scale screening of PM repair proteins using yeast GFP-tagged libraries and laser-induced injury assay in budding yeast. We identified 80 repair protein candidates, including 72 previously unreported candidates. Among the observed repair processes, we showed that polarized exocytosis and clathrin-mediated endocytosis (CME) are coupled at the damage site, with exocytosis predominating in the early stage of PM repair and CME predominating in the late stage of PM repair. We also showed that CME delivers repair proteins with transmembrane domains (TMDs) from the growing bud site (bud tip) to the damage site, presumably contributing to the restructuring of the damaged PM. These results provide insights into the temporal dynamics of coordinated cellular responses to PM damage.

Results

Proteome-scale identification of SDS-responsive proteins

To identify the PM repair proteins, we performed a two-step screening. First, we aimed to identify the proteins that change localization in response to SDS treatment, which induces local PM/cell wall damage to budding yeast (28). We used a yeast C-terminally GFP-tagged library (4159 ORFs) (29) and an N-terminally sfGFP-tagged library (N' Swat library) (5569 ORFs) (30, 31) comprising 86% of the yeast proteome (5718 ORFs in total) (Fig. 1A). We fixed the cells with paraformaldehyde after 1 hour of SDS treatment, in which condition the one of the known repair proteins, Pkc1-GFP, changes its localization (Fig. 1A). We successfully imaged the signal of 9181 proteins fused with GFP or sfGFP (5609 ORFs in total). We assessed the fluorescence signals of 5609 proteins in normal and SDS-treated conditions by visual inspection of the images. We identified 562 proteins whose fluorescence signal pattern changed after SDS treatment (Fig. 1B and Data S1). The hits included the localization changes of proteins, structural changes of organelles such as mitochondrial fragmentation, and foci/puncta formation in response to SDS treatment. In addition to Pkc1, we identified previously reported repair proteins, such as Rom2 and Dnf1, as screening hits (26, 27) (Data S1). Gene Ontology (GO) analysis of the screening hits revealed enrichment for proteins involved in the actin filament organization (Fig. 1C). This is consistent with the previous study that SDS treatment remodels the cell polarity and actin cytoskeleton (26). Based on the reported subcellular localization of proteins in normal growth conditions (29), we categorized the screening hits into six major classes and several minor ones representing less than ten proteins (Fig. 1D and Fig. S1, see Materials and Methods). We found that proteins that form puncta/foci after the SDS treatment include translation factor Gcd7-GFP (Fig. 1D), which forms foci in response to glucose deprivation (32). These observations suggest that SDS treatment induces stress responses, including those associated with PM/cell wall damage responses in budding yeast.

Laser-induced PM/cell wall damage assay identified 80 repair protein candidates

Our group previously developed a laser-induced PM/cell wall injury assay under live single-cell conditions (laser damage assay) (26, 33). We performed the laser damage assay to identify the PM repair proteins and observe their movements after PM/cell wall damage (Fig. 2A). The N-terminally sfGFP-tagged library is expressed under exogenous *NOP1* promoters, while the C-

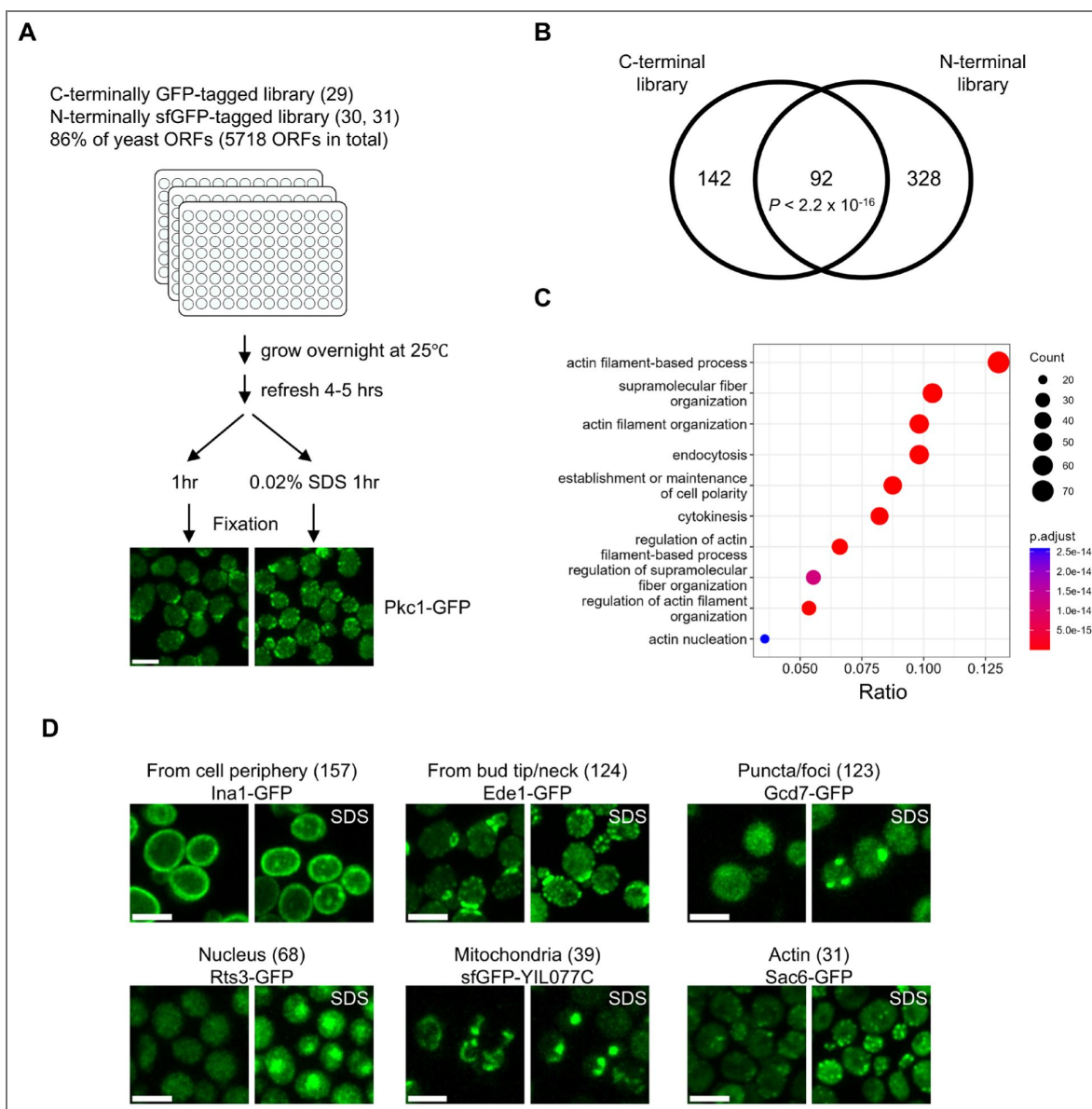


Fig. 1. Protein relocalization in response to SDS treatment.

(A) Schematic representation of screening methodology and images of Pkc1-GFP with or without 0.02 % SDS treatment are shown. Scale bar, 5 μ m. (B) Overlap of screening hits of C- and N-terminal libraries. P value for the significance of the overlap is indicated. Fisher's exact test was performed. (C) GO analysis of biological processes for the screening hits. Successfully observed proteins were used as the background protein sets. (D) Screening hits for six relocalization classes and the images of representative proteins in each class were shown. Numbers in parentheses indicate the number of proteins in the class. Scale bar, 2 μ m.

terminally GFP-tagged library is expressed under endogenous promoters (29, 30). To assess the proteins with endogenous expression levels, we selected the 234 screening hits from the C-terminal library as targets for the laser damage assay. During 25 min of observation at 30-second intervals, we identified 90 proteins whose localization changed in response to laser damage (Fig. 2B). We categorized the proteins into four classes based on their localization changes (Fig. 2C). Three of the puncta-forming proteins, Dna2, Dot6, and Gcd7, form puncta in response to cellular stresses (32, 34). The internalization of PM proteins (Dip5, Ina1, Ftr1, and Tpo1) is consistent with the fact that endocytosis of PM proteins contributes to cell adaptation to environmental stress (35). In addition, the transcription factors, Msn2 and Crz1, translocate from the cytoplasm to the nucleus after laser damage (Fig. 2B). Msn2 and Crz1 are activated under various stress conditions (36). Thus, the localization changes of puncta-forming proteins and transcriptional factors suggest common stress responses to laser damage and other cellular stresses. The most frequently observed localization changes were the accumulation at the damage site (Fig. 2B). We defined them as repair protein candidates, given their presumed involvement in the PM/cell wall repair process.

To gain further insight into the repair protein candidates, we classified them into four categories based on their subcellular localization under normal growth conditions (Fig. 2C). The largest category is bud-localized proteins (47 proteins) (Fig. 2C), which is consistent with a previous study showing that bud-localized proteins accumulate at the damage site in budding yeast (26). The repair protein candidates in this category include ten proteins that have transmembrane domains (TMDs), including phospholipid flippases (Dnf1/Dnf2) and osmosensor proteins (Slg1, Sho1) (Fig. 2C). Proteins with TMDs are transported to the PM as cargoes of secretory vesicles (37). Thus, this result suggests that secretory vesicles targeted to the damage site provide repair proteins with TMDs to the damage site. The second largest category is actin-localized proteins (25 proteins). This is consistent with the GO analysis that actin-binding proteins and proteins involved in actin-related biological processes are enriched in the repair protein candidates (Fig. 2, D and E). The proteins in this category include endocytic proteins, suggesting that endocytosis occurs at the damage site. Proteins that bind actin cables, such as Abp140, are included in this category, suggesting that actin cables are formed at the damage site. In addition, proteins involved in cell wall synthesis/maintenance (Flc1, Dfg5, Smi1, Skg1, Tos7, and Chs3) are identified as repair protein candidates (Data S2). These results are consistent with previous studies of PM/cell wall damage repair in budding yeast and human cells (11, 26, 38, 39).

The temporal order of the Pkc1 accumulation, polarized exocytosis, and CME at the damage site

To understand when repair proteins accumulate at the damage site, we defined their accumulation times as the time when the fluorescence intensity at the damage site exceeded the threefold standard deviation (3xSD) above the non-damaged site for at least two consecutive time frames (1 min). We also defined dispersion time when the fluorescence intensity at the damage site becomes less than that of the non-damaged site plus 3xSD for at least 1 min (Data S2).

The defined accumulation times raised the possibility that the Pkc1 pathway proteins accumulated at the damage site earlier than exocytosis regulators and exocytic cargoes with transmembrane domains (Dnf1, Dnf2, and Slg1) (Fig. S2). To verify this observation, we performed the laser damage assay in the cells expressing Exo70-mNeonGreen (mNG) and Pkc1-mScarlet-I (mSc-I), and the cells expressing Pkc1-sfGFP and Dnf1-mSc-I (Fig. 3A and B). Exo70 is one of the subunits of the exocyst complex, serving as a polarized exocytosis marker. The fluorescence-tagged Pkc1 accumulated at the damage site earlier than Exo70-mNG and Dnf1-mSc-I (Fig. 3, A and B). These results suggest that the Pkc1 accumulation at the damage site occurs earlier than the polarized exocytosis of transmembrane proteins.

Although not all exocyst subunits (eight subunits) were identified in this work, all the mSc-I-tagged exocyst subunits showed comparable fluorescence intensity changes with Exo70-mNG at the damage site and at the bud tip (Fig. S3, A to G). These results suggest that all subunits of the exocyst complex accumulate at the damage site simultaneously.

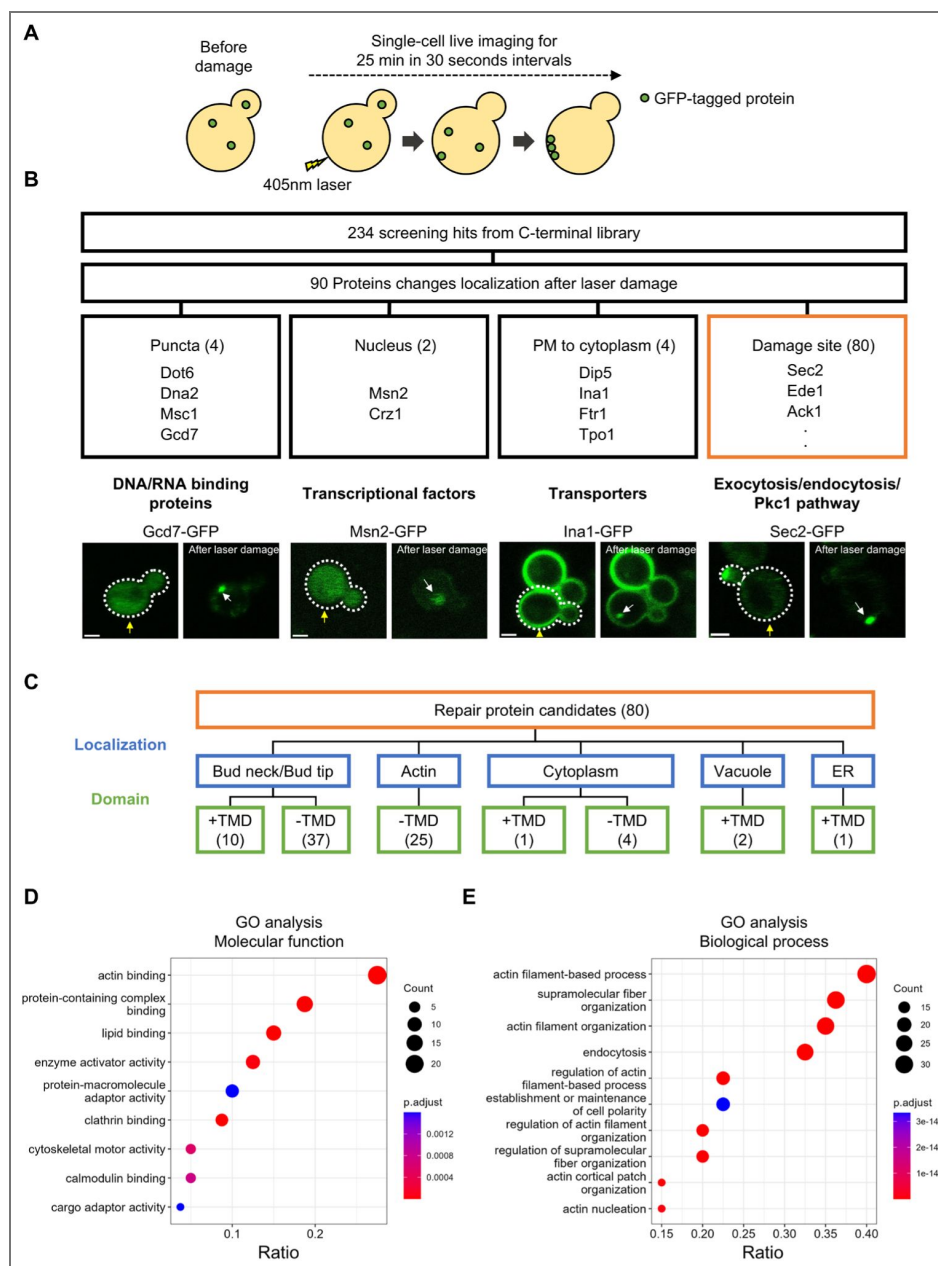


Fig. 2. Laser damage assays identified 80 repair protein candidates.

(A) Schematic representation of screening methodology. Cells were imaged for 25 min in 30-second intervals after 405 nm laser damage. (B) 90 proteins change localization after laser damage. The classification of the repair protein candidates based on their localization changes, representative proteins, and representative biological processes in each category is shown. Scale bar, 2 μ m. Yellow arrows show the damage site. White arrows show the recruitment of fluorescence signals. (C) Classification of repair protein candidates based on subcellular localization and domain. TMD+ represents the proteins that have transmembrane domains. The existence of transmembrane domains was predicted by TMHMM (63) (D) GO analysis of molecular functions for the repair protein candidates. The proteins whose localization changes in response to SDS treatment were used as background protein sets. (E) GO analysis of biological processes for the repair protein candidates. The proteins whose localization changes in response to SDS treatment were used as background protein sets. Ratio of the number of proteins associated with a specific GO term to the total number of proteins in the background is shown.

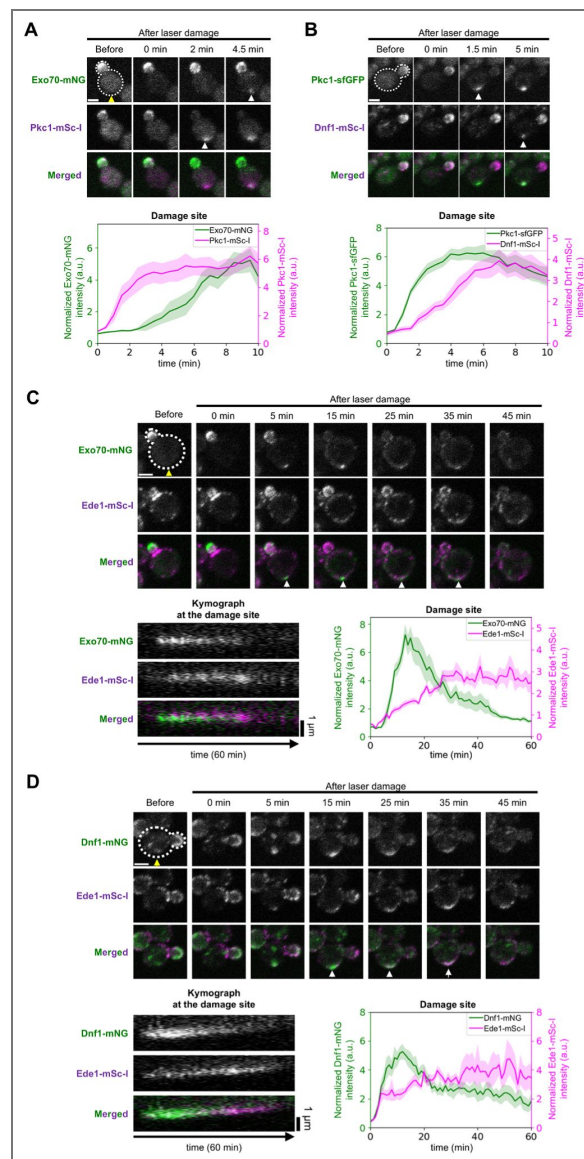


Fig. 3. The temporal order of the Pkc1 accumulation, polarized exocytosis, and CME at the damage site.

(A) The representative images and normalized fluorescence intensity of Exo70-mNG (Green) and Pkc1-mSc-I (Purple) at the damage site after laser damage. Yellow arrows show the damage site. White arrows show the recruitment of fluorescence signals. $n = 10$ cells. **(B)** The representative images and normalized fluorescence intensity of Pkc1-sfGFP (Green) and Dnf1-mSc-I (Purple) at the damage site after laser damage. Yellow arrows show the damage site. White arrows show the recruitment of fluorescence signals. $n = 8$ cells. **(C)** Representative images, kymograph at the damage site, and fluorescence intensity at the damage site of Exo70-mNG (Green) and Ede1-mSc-I (purple). $n = 10$ cells. Yellow arrows show the damage site. White arrows show the coaccumulation of Exo70-mNG and Ede1-mSc-I at the damage site. **(D)** Representative images, kymograph at the damage site, and fluorescence intensity at the damage site of Dnf1-mNG (Green) and Ede1-mSc-I (purple). $n = 9$ cells. Yellow arrows show the damage site. White arrows show the coaccumulation of Dnf1-mNG and Ede1-mSc-I. Lines and shaded regions are the mean and the standard error of the mean. White scale bar, 2 μm .

The CME markers, Sla1-GFP and Abp1-GFP (40), exhibited repeated short stays and reaccumulation at the damage site within 5 min to 25 min after laser damage (Fig. S4, A to C). These fluctuating movements of Sla1 and Abp1 are consistent with the previous studies (41, 42). Furthermore, Sla1-mNG and Abp1-mSc-I showed comparable fluctuating accumulation patterns in the same cell (Fig. S4D). These results suggest that CME repeatedly occurs at the damage site from around 5 min to 25 min after laser damage.

To understand the temporal order of polarized exocytosis and CME at the damage site, we performed the laser damage assay in the cells expressing Exo70-mNG/Dnf1-mNG and Ede1-mSc-I (Fig. 3, C and D). Ede1 is one of the earliest proteins recruited to endocytic sites, serving as a CME marker (43, 44). Exo70-mNG/Dnf1-mNG and Ede1-mSc-I accumulated at the damage site from 5 to 35 min (Fig. 3, C and D). The signals of Exo70-mNG/Dnf1-mNG peak within 20 min after the damage, while Ede1-mSc-I peaks 20 min after the damage (Fig. 3, C and D). These results suggest that CME and polarized exocytosis occur simultaneously at the damage site, but that polarized exocytosis predominates in the early stage of the PM/cell wall damage response, while CME predominates in the late stage of the PM/cell wall damage response.

Knockout mutants of CME are sensitive to PM/cell wall stresses

To identify the biological processes required for cell survival after PM/cell wall stress, we performed growth screening of repair protein knockout mutants in PM/cell wall stress conditions (Fig. S5A). We spotted the same number of yeast cells in YPD media, YPD+0.01% SDS media, YPD+25 µg/ml calcofluor white (CFW), and YPD media in 37 °C (Heat stress) and incubated for 3 days (Fig. S5A). CFW binds to chitin in the cell wall, inducing cell wall damage (45). Heat stress modifies PM structures (46). The growth assay showed that six knockout mutants of CME proteins, *rvs167Δ*, *sla1Δ*, *sla2Δ*, *end3Δ*, *las17Δ*, and *vrp1Δ*, are sensitive to all the stress conditions (Fig. S5B). We knocked out these genes using the WT strain in our laboratory and confirmed that these mutants are sensitive to SDS (Fig. S5C). These results led us to focus further on the CME functions associated with PM/cell wall damage repair.

CME proteins are required for polarized exocytosis at the damage site

Previous studies showed that endocytosis is involved in PM damage repair by directly resealing the PM by removing the damaged pores (47) or by restructuring the PM after resealing in mammalian cells (20, 21, 48). We showed that CME occurs at the damage site from 5 to 45 min after laser damage in budding yeast (Fig. 3, B and C). This supports the possibility that CME restructures the PM after resealing, which usually occurs within 1 min after the damage (20). Another potential mechanism for restructuring the damaged PM is exocytosis, which delivers PM proteins and lipids to the damage site. Our previous work showed that the polarized exocytosis machinery is directed from the bud tip to the damage site in response to laser damage in budding yeast (26). Given that CME and polarized exocytosis constitute a coupled transport cycle in budding yeast (49), we reasoned that CME positively regulates polarized exocytosis at the damage site, thereby restructuring the damaged PM.

To test this idea, we performed the laser damage assay in SDS-sensitive endocytic mutants (*sla1Δ*, *end3Δ*, *vrp1Δ*, and *rvs167Δ*). We used two exocytosis markers, type V myosin (Myo2) and the exocyst subunit (Exo70), fused to sfGFP and mNG, to assess the exocytosis activity at the damage site. The effect of deleting the CME genes on the expression levels of Myo2-sfGFP and Exo70-mNG before laser damage was minimal (Fig. S6, A and B). The accumulation of Myo2-sfGFP and Exo70-mNG at the damage site was impaired in *sla1Δ*, *end3Δ*, and *vrp1Δ* (Fig. 4, A and B, fig. S6, A and B). Moreover, we also found that the dispersion of Myo2-sfGFP and Exo70-mNG from the bud tip was partially inhibited in CME mutants (Fig. 4, A and B, fig. S6, A and B). The observed weak phenotype of *rvs167Δ* for the accumulation and dispersion of Myo2-sfGFP and Exo70-mNG is consistent with the previous study that exocytosis is not impaired in *rvs167Δ* (49).

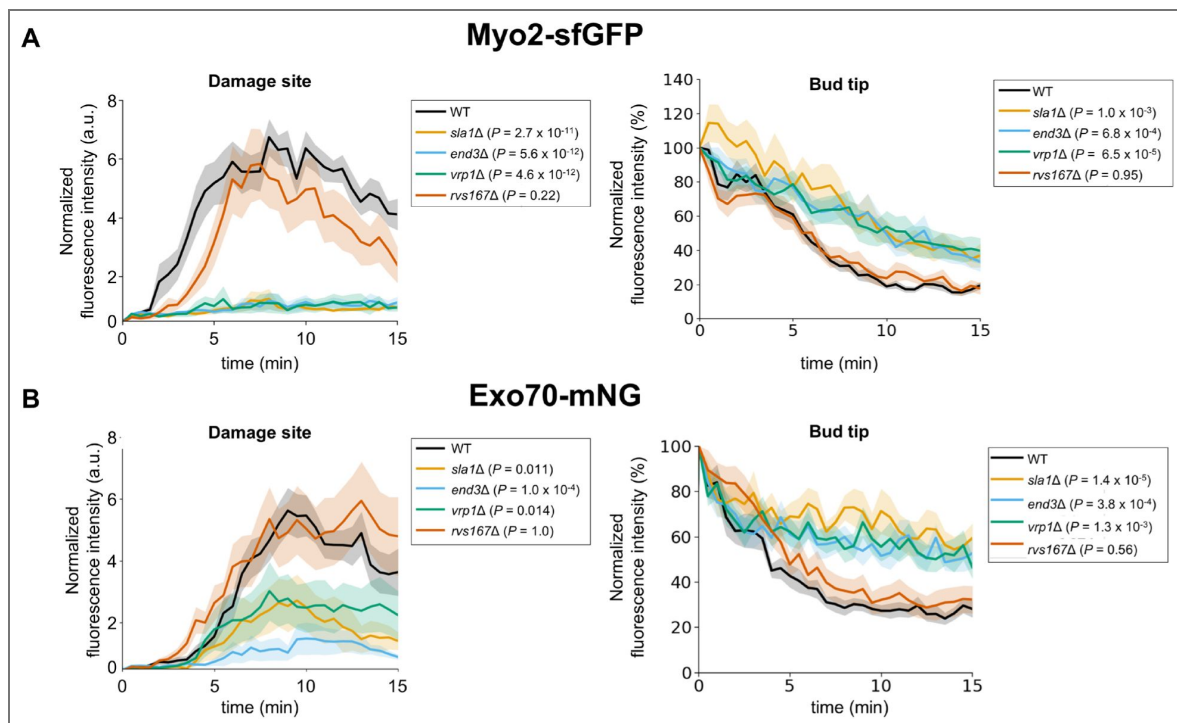


Fig. 4. CME proteins are required for polarized exocytosis at the damage site.

(A) The quantification results of fluorescence intensity of Myo2-sfGFP at the damage site and at the bud tip. $n = 13$ for WT, $n = 12$ for $end3\Delta$, $sla1\Delta$, $rvs167\Delta$, and $vrp1\Delta$. (B) The quantification results of the fluorescence intensity of Exo70-mNG at the damage site. $n = 18$ for WT, $n = 12$ for $end3\Delta$, $sla1\Delta$, and $rvs167\Delta$, $n = 14$ for $vrp1\Delta$. Lines and shaded regions are the mean and standard error of the mean. The maximum fluorescence intensity at the damage site or fluorescence intensity changes at the bud tip were compared between the WT and each CME mutant using Dunnett's multiple comparison test.

Our previous work also showed that the actin nucleator formin Bni1 and the exocyst subunit Sec3 are degraded in response to SDS treatment (26). To test the possibility that CME is required for the decrease of Bni1 and Sec3 after SDS treatment, we performed immunoblotting of Bni1-13xMyc and Sec3-GFP before and after SDS treatment (Fig. S7, A and B). Protein levels of Bni1-13xMyc and Sec3-GFP were decreased in WT, *rvs167Δ*, and *end3Δ* after SDS treatment (Fig. S7, A and B). Given that *end3Δ* impaired the accumulation of Myo2-sfGFP and Exo70-mNG at the damage site (Fig. 4, A and B), these results suggest that CME is dispensable for the decrease of Bni1 and Sec3 after SDS treatment. Altogether, these results suggest that CME is involved in the direction of the Exo70-mNG and Myo2-sfGFP to the damage site in a Bni1- and Sec3-degradation-independent manner.

CME at the bud tip directs repair proteins with TMDs to the damage site

Although the requirement of Rvs167 for the Pkc1 accumulation and polarized exocytosis at the damage site is minimal (Fig. 4, A to C), *rvs167Δ* is sensitive to SDS (Fig. S5C). This raised the possibility that CME plays additional roles, such as membrane protein trafficking, in PM repair. We found that CME proteins, including Apl1, Ede1, and Ent1, changed their localization from the bud neck to the bud tip within 10 min after laser damage (Fig. S8A). The fluorescence intensity changes of CME proteins at the bud tip at 10 min after laser damage were higher than those of other repair protein candidates (Fig. S8B). Consistent with these results, the Ede1-mSc-I signal at the bud tip increased within 10 min after laser damage, and then the Ede1-mSc-I signal at the damage site increased (Fig. 5A). These results suggest that CME occurs at the bud tip within 10 min after laser damage. We also found that Dnf1-mNG disappeared from the bud tip following the recruitment of Ede1-mSc-I to the bud tip (Fig. 5, A and B). The disappearance from the bud tip and the accumulation at the damage site of Dnf1-mNG were impaired in *rvs167Δ* (Fig. 5, B and C, and fig. S9). Furthermore, the disappearance from the bud tip and the accumulation at the damage site of repair proteins with TMDs (Slg1-sfGFP and Sho1-GFP) and representative endocytic recycling cargo, mNG-Snc1, were impaired in *rvs167Δ* (Fig. 5, D to F and fig. S10, A to C). These results support our idea that CME directs repair proteins with TMDs from the bud tip to the damage site.

mNG-Snc1 is retargeted from the damage site to the bud tip

The fluorescence intensity of accumulated repair proteins with TMDs at the damage site gradually decreases approximately 15 min after laser damage (Fig. 3D and fig. S10, A and C). Using mNG-Snc1 as a model, we investigated the destination of repair proteins after PM repair. To avoid observing the mNG-Snc1 that is newly synthesized after laser damage, we transiently expressed mNG-Snc1 under the control of the galactose-inducible promoter (Gal1pr) before laser damage (Fig. 6A). After 1 hour of mNG-Snc1 expression in galactose media, we transferred the cells to glucose media to stop the expression (Fig. 6A). To minimize the effect of changing the carbon source, we further incubated the cells in glucose media for at least 3 hours prior to laser damage (Fig. 6A). The mNG-Snc1 accumulated at the damage site approximately ~10 min after laser damage, gradually disappearing after the colocalization with Ede1-mSc-I (Fig. 6A and B). The mNG-Snc1 signal at the bud tip decreased to 10-53% 22 min after laser damage (Fig. 6C and D). The normalized mNG-Snc1 signal at the bud tip recovered to 54-88% 50 min after laser damage. (Fig. 6, C and D). These results suggest that mNG-Snc1 is redirected to the bud tip after PM repair.

Discussion

There has been a growing interest in the mechanisms underlying PM repair, partly due to their association with human diseases and cellular aging (3-5, 28). PM repair proteins, which accumulate at the damage site, play critical roles in PM repair. In this study, by large-scale identification of PM repair proteins and single- and dual-color live-cell imaging of repair proteins, we analyzed spatiotemporal PM repair processes in budding yeast. We propose a model in which

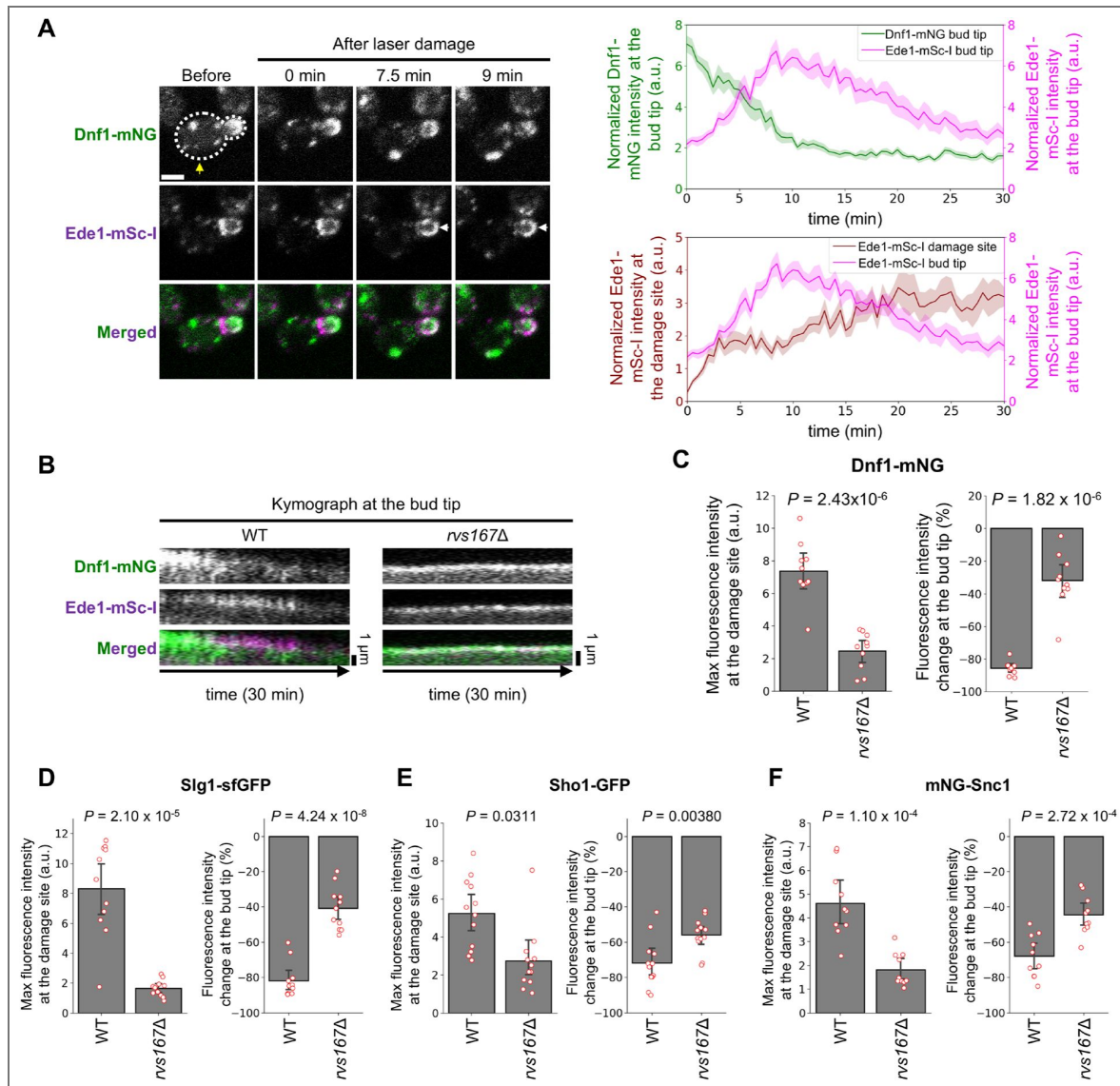


Fig. 5. CME at the bud tip directs repair proteins with TMDs to the damage site.

(A) Representative images and the normalized fluorescence intensity at the bud tip of Dnf1-mNG and Ede1-mSc-I. Yellow arrows show the damage site. White arrows show the recruitment of fluorescence signals. Scale bar, 2 μ m. (B) Kymograph of Dnf1-mNG and Ede1-mSc-I at the bud tip in WT and *nvs167Δ*. $n = 10$ cells. (C)–(F) Max fluorescence intensity at the damage site and fluorescence intensity changes at the bud tip in WT and *nvs167Δ*. $n = 10$ cells for Dnf1-mNG. $n = 11$ cells for Slg1-sfGFP. $n = 12$ cells for Sho1-GFP. $n = 10$ cells for mNG-Snc1. Welch's t-test was performed.

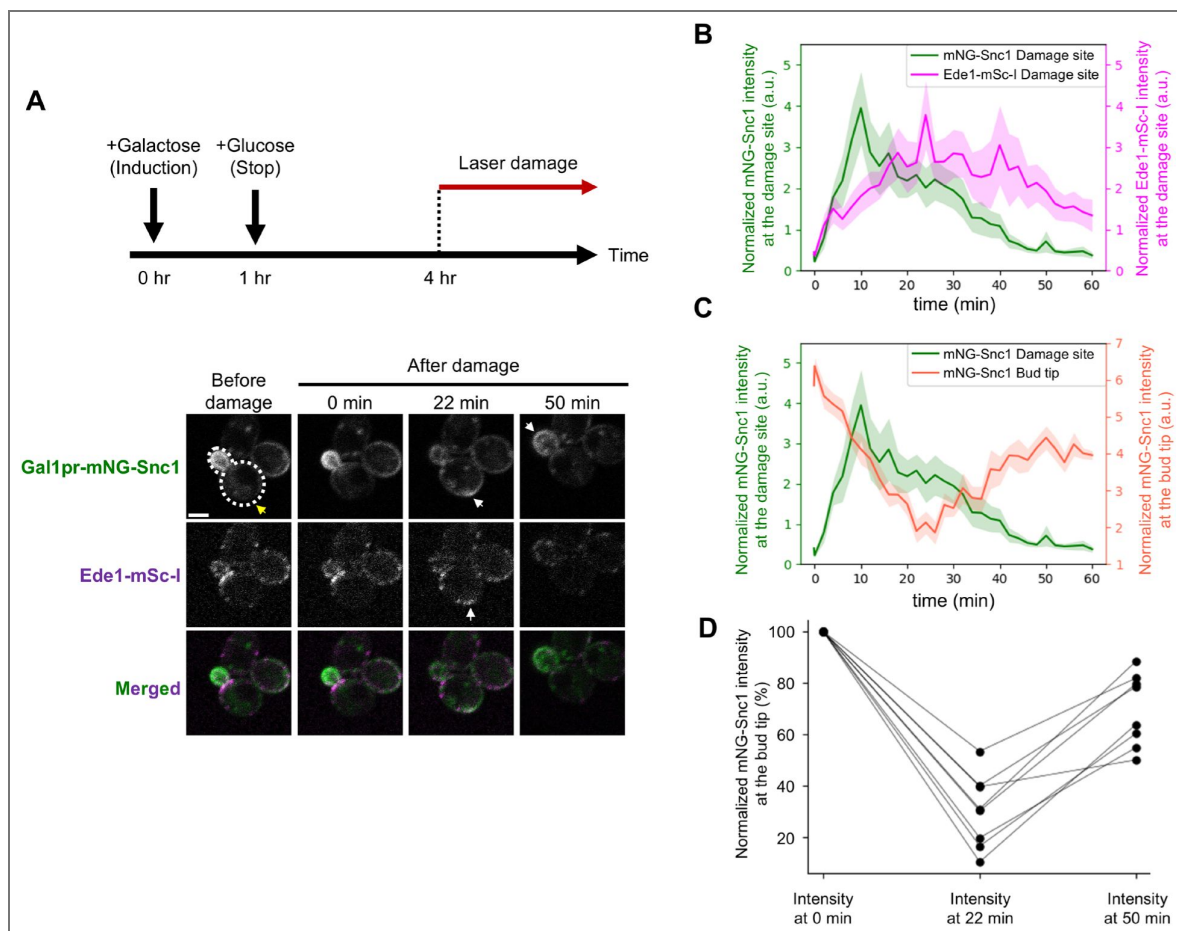


Fig. 6. mNG-Snc1 is recovered from the damage site to the bud tip after PM repair.

(A) Schematic of transient expression induction of mNG-Snc1 by Gal1 promoter and representative images of mNG-Snc1 and Ede1-mSc-I. After transcription activation of the Gal1 promoter by adding 3% galactose, we stop the expression by transferring the cells to glucose media. The cells were incubated for at least 3 hours before the laser damage assay. Yellow arrows show the damage site. White arrows showed the recruitment of fluorescence signals. Scale bar, 2 μ m. **(B)** Quantification of mNG-Snc1 (Green) and Ede1-mSc-I (purple) at the damage site. **(C)** Quantification of mNG-Snc1 at the bud tip (Green) and at the damage site (tomato). **(D)** The changes in the normalized mNG-Snc1 signal at the bud tip. $n = 8$ cells.

CME at the bud tip and at the damage site delivers repair proteins with TMDs between the bud tip and the damage site, allowing the cell to restructure the damaged PM and to resume growth after PM repair (Fig. 7 [↗](#)).

Two-step visual screening for PM repair protein identification

By combining proteome-scale visual screening using yeast GFP libraries and the laser damage assay, we identified 80 repair protein candidates (Fig. 2B [↗](#) and Data S2). Strikingly, 72 out of 80 repair protein candidates were not previously reported to accumulate at the damage site in budding yeast ([26, 27](#)). The unreported repair protein candidates include uncharacterized proteins, such as Sap1 and Ypr089w (Data S2). Characterizing these proteins may expand our understanding of PM repair mechanisms.

We selected the screening hits from the C-terminally GFP-tagged library as targets for the laser damage assay. Although this is a reasonable approach to evaluating the proteins from the endogenous expression level, it overlooks the potential repair proteins in the N-terminally sfGFP-tagged library. Around 23% of ORFs only exist in the N-terminal library, and 11% of yeast ORFs show different localization from that of the C-terminal library ([31](#)). For example, sfGFP-Snc2, but not Snc2-GFP, changes localization in response to SDS treatment because of the mislocalization of Snc2-GFP in the vacuole. Because Snc2 is a homolog of Snc1, Snc2 is a potential repair protein. In addition, 48 out of 80 repair proteins identified in this work were hits in the N-terminally sfGFP-tagged library screening (Table S1 and S2). Screening hits from the N-terminally sfGFP-tagged library can also be a useful resource for further identification of repair proteins.

The screening is unable to identify proteins whose localization remains unaltered in response to SDS treatment. For example, we could not identify Cdc50-Drs2, which accumulates at the damage site induced by a laser ([27](#)). A recent study demonstrated the different cellular responses between focal (laser damage) and diffuse (streptolysin-O (SLO) treatment) PM damage ([6](#)). Some proteins may be overlooked in the screening due to the intrinsic differences between SDS treatment and the laser damage assay. Using different stresses to induce PM damage may identify additional PM repair proteins. In addition, the screening did not identify ESCRT proteins as hits. The tagging of fluorescent proteins occasionally interferes with the functions of the tagged proteins ([50](#)). Specifically, the exogenous expression of fluorescent-tagged ESCRT subunits can impair their functions ([51, 52](#)). The screening may have overlooked some of these proteins, possibly including ESCRT proteins.

We performed live-cell imaging of repair proteins and defined accumulation times of repair proteins (Fig. S2 [↗](#) and Data S2). This dataset provides an overview of repair protein accumulation. However, it should be noted that the accurate relative accumulation timing of repair proteins should be determined by multi-color imaging of repair proteins in the same cells, because of cell-cell variabilities, the GFP-tagging effects on the cells, and the small sample size of the screening. These datasets will form the basis for future hypothesis-driven studies.

Coordination of polarized exocytosis and CME

We previously showed that formin Bnr1 forms the actin cable at the damage site ([26](#)). Consistent with this, we identified Bnr1-interacting proteins, including Bud6 and Smy1, as repair proteins (Data S2). Because actin cables mediate polarized exocytosis and secretory vesicle trafficking ([53](#)), these results further support our model that exocytosis delivers repair proteins with TMD at the damage site (Fig. 7 [↗](#)).

We showed that polarized exocytosis and CME occur simultaneously at the damage site between approximately 5 and 35 min after laser damage (Fig. 3, C and D [↗](#)). Given that the resealing of the damaged PM generally occurs within 1 min ([20](#)), this result implies that polarized exocytosis and CME are involved in the restructuring of the damaged PM rather than the resealing of it. The coupling of endocytosis with polarized exocytosis is observed in multiple cell types, including at the growing tip of budding yeast, the growing tip of pollen tubes, and synapses in neurons ([54](#)). Polarized exocytosis and CME at the damage site may regulate the PM tensions and amount of lipids and PM proteins in a manner analogous to these cells.

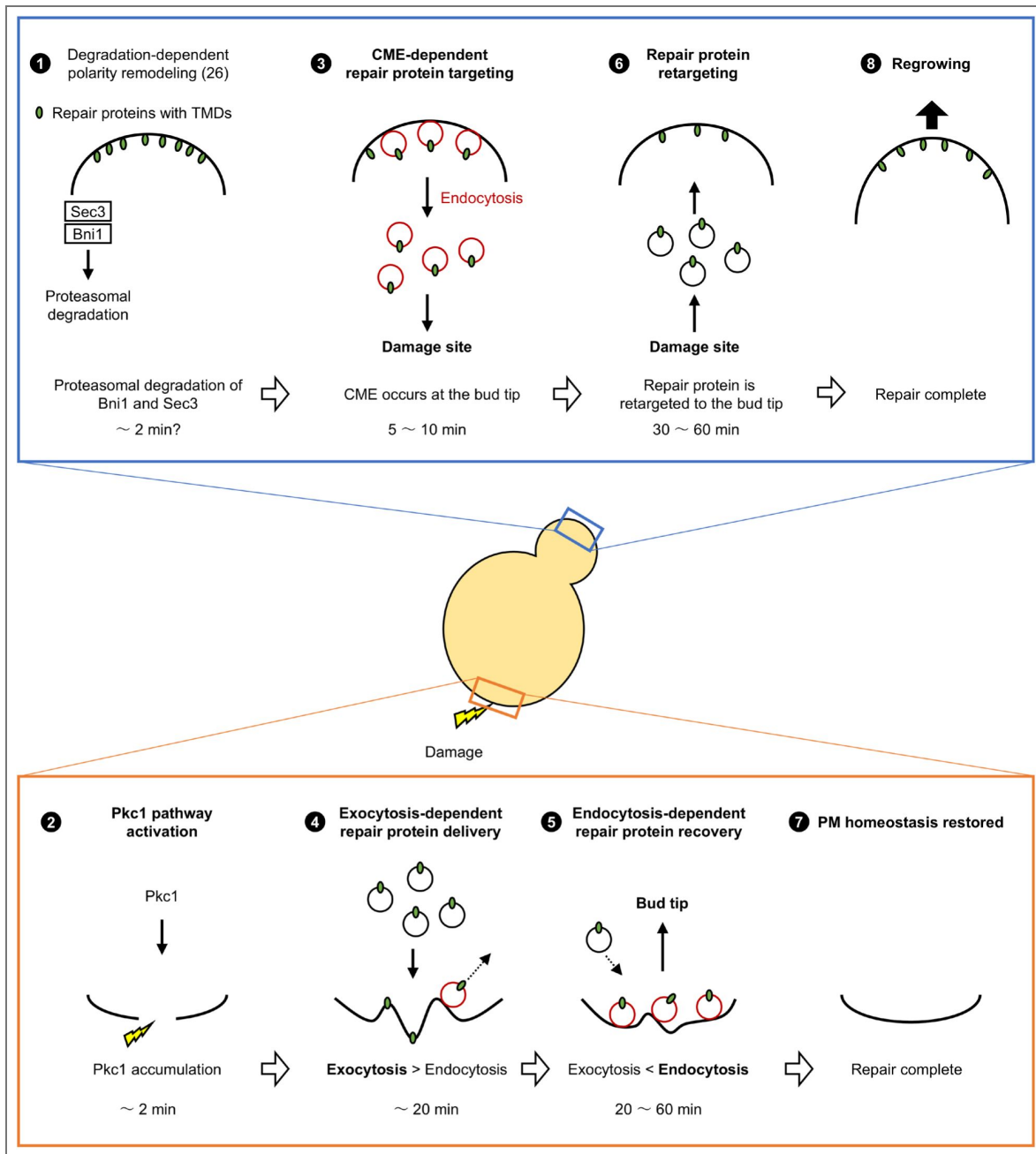


Fig. 7. Model of spatiotemporal cellular responses to PM damage in budding yeast.

We showed the hypothetical model of spatiotemporal PM damage responses in budding yeast. First, the degradation of Sec3 and Bni1 resolved the polarity competition between the bud tip and the damage site (26). Within 10 min after laser damage, CME directs repair proteins with TMDs to the damage site from the bud tip. At the damage site, polarized exocytosis and CME simultaneously occur, with exocytosis predominating approximately within 20 min and with CME predominating approximately 20 min after laser damage. CME targets repair proteins with TMDs from the bud tip to the damage site. The endocytosed PM proteins are retargeted to the bud tip again after the PM repair is finished. The retargeted PM proteins may be involved in the resumption of cell growth after PM repair. The numbers represent the temporal order of events.

Our results suggest that the activity of CME and polarized exocytosis at the damage site changes over time, with exocytosis predominating within 20 min after laser damage and CME predominating 20 min after laser damage (Fig. 3, C and D [↗](#)). This is consistent with the recent study in human cells (21). A previous study showed that polarized exocytosis activates CME in budding yeast (49). Rab GTPase Sec4, which regulates polarized exocytosis, also activates endocytosis by overriding Sla1 inhibition of Las17 (49). We could not identify Sec4 as a repair protein candidate because C-terminally GFP-tagged Sec4 is not in the library, probably due to the loss of function of C-terminally tagged Sec4. However, we found that Sec2-GFP, the guanine nucleotide exchange factor (GEF) of Sec4, accumulates at the damage site (Data S2). Sec4 and Sec2 are localized to the secretory vesicles (55, 56). Therefore, secretory vesicles that are targeted to the damage site may accumulate Sec2 and Sec4, leading to the activation of CME. The switching of activities between polarized exocytosis and CME may contribute to restoring PM homeostasis after the damage.

Polarized exocytosis at the damage site is inhibited in CME mutants (Fig. 4, A and B [↗](#), and fig. S6, A and B [↗](#)). Given that CME and polarized exocytosis occur simultaneously (Fig. 3, C and D [↗](#)), CME may activate polarized exocytosis at the damage site, such as by regulating the PM tension (57) around the damage site. Moreover, Myo2-sfGFP and Exo70-mNG were partially retained at the bud tip after laser damage in CME mutants (Fig. 4, A and B [↗](#), and fig. S6, A and B [↗](#)). These results raise the possibility that CME is involved in targeting Exo70 and Myo2 from the bud tip. At the bud tip, CME may be involved in the dispersion of Exo70 and Myo2 via upstream regulators, such as Rho3.

CME functions for PM repair in budding yeast

Previous studies showed that endocytosis actively occurs at the damage site to repair the damaged PM (11, 12, 20). Surprisingly, our results suggest that CME occurs not only at the damage site but at the bud tip within 10 min after laser damage. In *rvs167Δ*, in which exocytosis inhibition at the damage site is minimal (Fig. 4, B and C [↗](#)), repair proteins with TMDs at the bud tip fail to accumulate at the damage site (Fig. 5, B to F [↗](#)). These results are consistent with our idea that CME at the bud tip directs repair proteins with TMDs to the damage site (Fig. 7 [↗](#)). Consistent with our results, dysferlin-containing vesicles increase in response to PM damage in muscle cells (25, 58). Dysferlin has a TMD and functions as a PM repair protein (25, 58, 59). Delivery of repair proteins with TMDs from non-damaged sites to the damage site by endocytosis may occur in a wide range of eukaryotic species.

We showed that 54–88% of the normalized mNG-Snc1 signal at the bud tip is recovered 50 min after the damage (Fig. 6, C and D [↗](#)). Because Snc1 is a CME cargo and it colocalizes with Ede1-mSc-I at the damage site, it is presumably recovered from the damage site via CME. Previous studies showed that macropinocytosis and CME restructure the damaged PM in human cells (20, 21). However, it is not clear whether the cargo proteins are recycled or degraded. We propose that CME terminates the PM damage responses by removing the repair proteins with TMDs from the damage site and resuming the growth by retargeting them to the bud tip (Fig. 7 [↗](#)). Further confirmation of the proposed model requires other experimental strategies, such as the photo-bleaching and photo-conversion of repair proteins, and the use of other repair proteins with TMDs.

Here, we showed spatiotemporal cellular responses after PM damage in budding yeast by large-scale identification of repair proteins and their live-cell imaging. Despite the limitations mentioned above, our datasets provide the first functional catalog for PM repair proteins. Because some of the PM repair proteins identified in this study and CME mechanisms are evolutionarily conserved, this work may serve as a basis for future studies, including those to be conducted in mammalian cells.

Materials and Methods

Media and strains

Standard procedures were used for DNA, *E. coli*, and yeast genetic manipulation. Yeast transformations were performed using the lithium acetate method. A PCR-based procedure was used for gene deletion. The deletion of the expected locus was confirmed by colony PCR (60). Yeast cells were cultured in YPD media (1% yeast extract, 2% bacto peptone, and 2% glucose) unless otherwise indicated. SD media (yeast nitrogen base (6.7 g/liter) without amino acids, L-adenine (550 mg/liter), L-arginine (280 mg/liter), L-alanine (280 mg/liter), L-asparagine (280 mg/liter), L-aspartic acids (280 mg/liter), L-cysteine (280 mg/liter), glycine (280 mg/liter), L-glutamic acids (280 mg/liter), L-glutamine (280 mg/liter), L-isoleucine (280 mg/liter), L-lysine (280 mg/liter), L-phenylalanine (280 mg/liter), L-proline (280 mg/liter), L-serine (280 mg/liter), L-threonine (280 mg/liter), L-tyrosine (280 mg/liter), L-valine (280 mg/liter), leucine (530 mg/liter), methionine (86 mg/liter), histidine (86 mg/liter), uracil (22 mg/liter), myo-inositol (100 mg/liter), and *p*-aminobenzoic acid (3 mg/liter) (pH 5.5)) were used for the laser damage assay. Yeast culture was performed at 25°C unless otherwise indicated. The yeast strains, yeast libraries, and plasmids used in this study are listed in Tables S1 to S3.

For the mNG-Snc1 expression experiments in Fig. 6, we grow cells in SD media containing 2% raffinose instead of glucose overnight. Then, we induced the expression of mNG-Snc1 by adding 3% galactose. After one hour of growth in galactose media, the cells were centrifuged at 5000xg for 5 min, and the spun-down cells were washed twice with SD media containing 2% glucose. We transferred the spun-down cells to SD media containing 2% glucose. The cells were incubated for at least 3 hours before the laser damage assay.

GFP screen for SDS damage response

GFP strains were spotted onto the YPD plates from 96-well plates using a pin replicator and incubated at 25°C. After 3 days, the colonies were inoculated into 200 µl of YPD media and incubated overnight to saturation. After mixing, 3 µl of the saturated culture was transferred to 1 ml of YPD media in deep well plates and incubated for 4-5 hours. 500 µl of the cultures were transferred to the new deep well plates. 10 µl of 1% SDS was added to 500 µl of the cultures and incubated at 25°C for 1 hour. The cells were centrifuged at 2000xg for 2 min. The cells were fixed by 300 µl of 4% PFA in YPD for 30 min at room temperature. The fixed cells were centrifuged at 2000xg for 2 min. Cells were washed twice with PBS and centrifuged at 2000xg for 2 min. The supernatant was removed between each wash. Cells were kept at 4°C until imaging. Cells were imaged with an LSM880 confocal microscope using a 20x air objective lens with an airyscan detector for GFP fluorescence (Ex 488/Em522). Maximum intensity projections of z-stack images are shown.

Categorization of localization changes in response to SDS treatment

We manually reviewed the acquired images using Zen Blue edition (Zeiss) or Fiji software (61) and compared the fluorescence signals of GFP-tagged proteins in normal and SDS treatment conditions. We identified the proteins whose fluorescence signal pattern changes between the two conditions. We categorized the screening hits based on reported protein localization (29). We first categorized the bud tip- or the bud neck-localizing proteins as “From bud tip/neck”. Among the uncategorized remaining screening hits, the actin-localizing proteins were categorized as “Actin”. Among the remaining uncategorized screening hits, cell periphery-localizing proteins were categorized as “From cell periphery.” Among the remaining uncategorized screening hits, nucleus-localizing proteins were categorized as “Nucleus” or “Nucleus to the cytoplasm”. We categorized Urc1 as “Nucleus to cytoplasm” because its signal dispersed to the cytoplasm in an SDS treatment condition. Other proteins in the “Nucleus” showed stronger nucleus localization in the SDS treatment conditions. Among the remaining uncategorized screening hits, we categorized

mitochondria-localizing proteins as “Mitochondria”. Among the remaining uncategorized screening hits, we categorized spindle pole-localizing proteins as “Spindle pole.” All proteins in the “Spindle pole” showed stronger dot-like structures in the SDS treatment condition. Among the remaining uncategorized screening hits, we categorized proteins that show puncta or foci in SDS treatment conditions as “Puncta/foci.” We categorized Gcn2, Ato3, Sun4, Csi2, Sps4, and Ypr089w as “Puncta/foci dispersed” because they showed weaker foci or puncta signals in SDS treatment conditions. We categorized Tul1, Emp24, Hor7, Scw10, and Ynl019c as “From vacuole” because they showed decreased vacuole signal in the SDS treatment conditions. We categorized Prm5 as “to vacuole” because its vacuole signal increases. We categorized Yps1 and Msc1 as “cytoplasm to ER” because they localize to the ER, and their ER signal increased in SDS treatment conditions. In this study, we make use of the protein localization data from the *Saccharomyces* Genome Database (SGD) and (29). All results of this screening are shown in Data S1.

Laser damage assay

Yeast cells were grown in SD medium at 25°C until the culture reached an OD600 of 0.1–0.4. We diluted the yeast culture and further incubated the yeast cells in SD medium for 3–8 hours at 25°C. We took 1 mL of the culture and centrifuged it at 500xg for 5 min to spin the cells down. We took 5–10 µL of the cell suspension and placed it onto SD medium + 2.2% agarose bed. A Concanavalin A (ConA) (Nacalai Tesque) coated glass slip was placed on the cells prior to imaging.

Cells were observed with A1R (Nikon). A1R was equipped with a CFI Apochromat TIRF 60×/1.49 oil objective lens (Nikon). GFP, mNG, and sfGFP were excited by a 488 nm laser, and the fluorescence that passed a 525/50 nm band-pass filter was detected with a GaAsP detector. mSc-I was excited by a 561 nm laser, and the fluorescence that passed a 595/50 nm band-pass filter was detected with a GaAsP detector. For the laser damage assay, the 405 nm laser was irradiated to a circle of 0.5 µm diameter in a cell periphery. The laser power was set between 30% and 70%, depending on the fluorescence-tagged proteins. We determined the laser power sufficient for repair protein recruitment at the damage site in control cells without resulting in cell lysis during the experiments. At least ~80% of cells are viable during the experiments, otherwise indicated. Only the cells that survive after laser damage are quantified. To compare the accumulation of repair proteins between different strains, we set the exact same laser power and microscopy.

Categorization of localization changes in response to laser damage

We categorized proteins that accumulate at the damage site as “Damage site”. These proteins are defined as repair protein candidates. Among the remaining uncategorized proteins, we categorized cell periphery-localizing proteins as “PM to cytoplasm”. Among the remaining uncategorized proteins, we categorized Msn2 and Crz1 as “Nucleus” because they changed localization from the cytoplasm to the nucleus. Among the remaining uncategorized proteins, we categorized Dot6, Dna2, Msc1, and Gcd7 as “Puncta” because they formed punctate structures in response to laser damage. The repair protein candidates identified in this study are listed in Data S2.

Quantification of fluorescence signal for laser damage assay

The quantification of the fluorescence signal from the laser damage assay was performed as described previously (27). For the large-scale identification of repair protein candidates, we selected the region of interest (ROI) around the whole cell, the damage site, and the bud tip of the cells. We manually move the ROIs as the cell moves so that the ROIs remain at the same position in the cell. We set the ROIs and manually selected them for quantification. The fluorescence intensity at the damage site and the bud tip were normalized by the fluorescence intensity of a whole cell to minimize the effect of photobleaching during imaging.

We defined the accumulation time as the time when the fluorescence intensity at the damaged site became greater than the fluorescence intensity at the non-damaged site by three times the standard deviation (3xSD) for at least two consecutive time frames. We defined the dispersion time when the fluorescence intensity at the damage site becomes less than that of the non-damaged site

plus 3xSD for at least two consecutive time frames after the accumulation time. We defined the retention time as the difference between the accumulation time and the dispersion time. We measured at least three cells. The median values of accumulation times and retention times across replicates for all repair proteins are listed in Data S2.

Growth screening of repair protein knockout mutants

Yeast cultures grown overnight in YPD at 25°C were diluted to OD600 = 0.1. The diluted cultures were spotted onto YPD, YPD + 0.01% SDS, and YPD + 25 µg/ml CFW plates. After 3 days of incubation at 25°C or 37°C (Heat stress), the growth of the YPD plate and other plates was compared. We performed the screening of two independent colonies. The strains were from the yeast deletion collection library (62) except for *smi1Δ*, *las17Δ*, *skg6Δ*, *end3Δ*, *vrp1Δ*, and *sla2Δ*. Only the mutants that showed sensitivity to the stress in two independent colonies were defined as sensitive to the stress. The results of the screening are shown in Data S2.

Spot assay

Yeast cultures grown overnight in YPD at 25°C were diluted to OD600 = 0.1. The four-fold serial dilutions of cultures were spotted onto the indicated plates.

Immunoblotting

Yeast cells in the early log phase (OD600 = 0.1–0.3) were treated with 0.02% SDS. 5 ml of yeast cells were collected before SDS treatment, 1 hr after SDS treatment, and 2 hr after SDS treatment. Cells were flash frozen by liquid nitrogen and stored at –80°C. Cells were resuspended in 120 µl of cold lysis buffer (0.25 M NaOH and 1% β-mercaptoethanol) and incubated on ice for 10 min. 20 µl of trichloroacetic acid was added to the lysates. After 10 min of incubation on ice, the lysates were spun down, and the supernatant was discarded. The precipitates were washed with 500 µl ice-cold acetone twice and dried at room temperature. The precipitates are resuspended in SDS polyacrylamide gel electrophoresis (PAGE) sample buffer [63 mM Tris-Cl (pH 6.8), 2% SDS, 1% β-mercaptoethanol, 0.01% Bromophenol Blue, and 10% glycerol]. Immunoblotting was performed with anti-mini c-Myc (SANTA CRUZ, sc-40), anti-GFP (Roche Merck, 11814460001), or anti-tubulin α (Bio-Rad, MCA78G) antibodies.

Statistical Analysis

Dunnett's multiple comparison test, Welch's t-test, and Mann–Whitney U test were performed with Python software (v. 3.11). Gene enrichment analysis and Fisher's exact test were performed with R software (v. 4.2.2).

Supplementary Materials

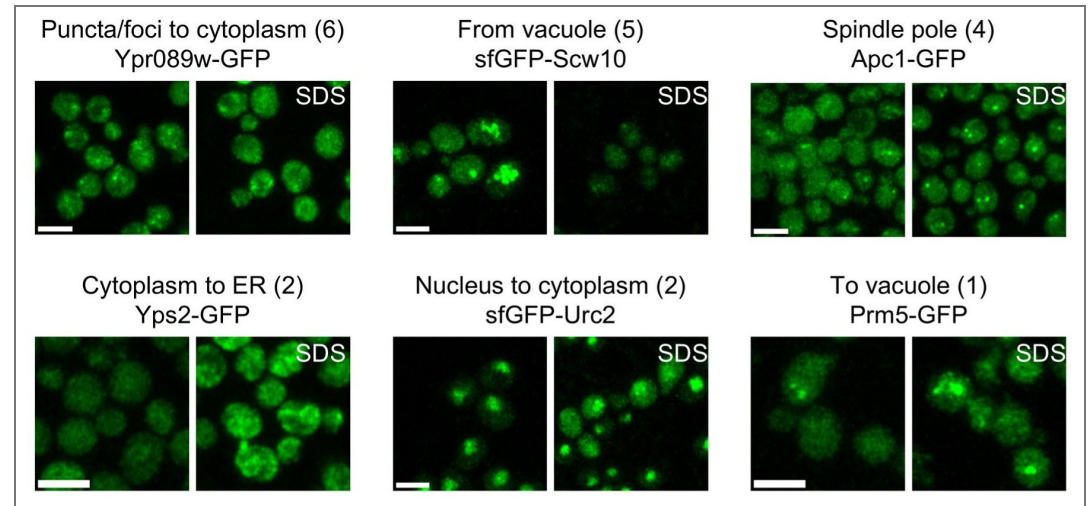


Fig. S1. Minor classes of localization changes in response to SDS treatment. Representative images of GFP- or sfGFP-tagged proteins in minor classes with or without 0.02% SDS treatment are shown. Numbers in parentheses indicate the number of proteins in the class. Scale bar 2 μ m.

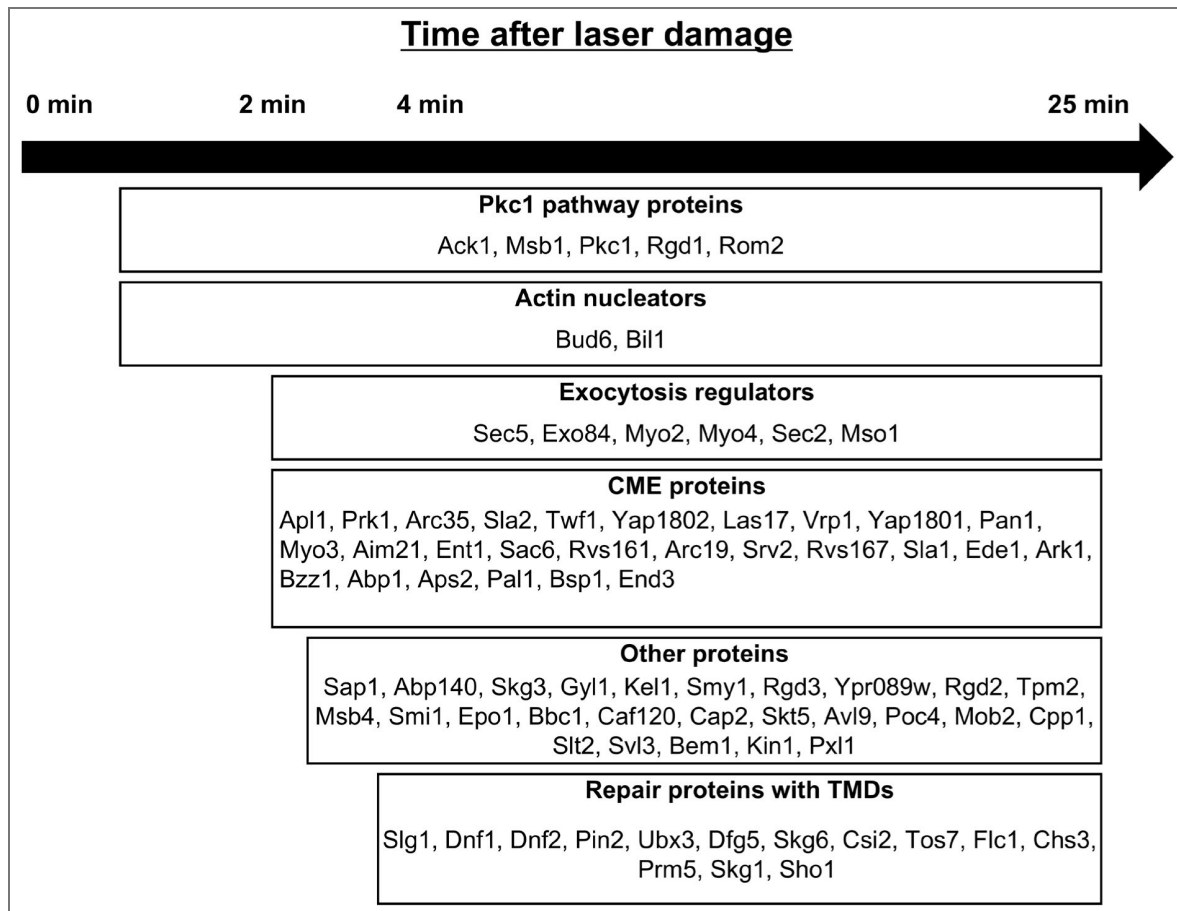


Fig. S2. Summary of the accumulation time of repair proteins.

The repair protein candidates were grouped based on their functions. The box ranges from the accumulation times to the dispersion times of grouped proteins. See also Data S2.

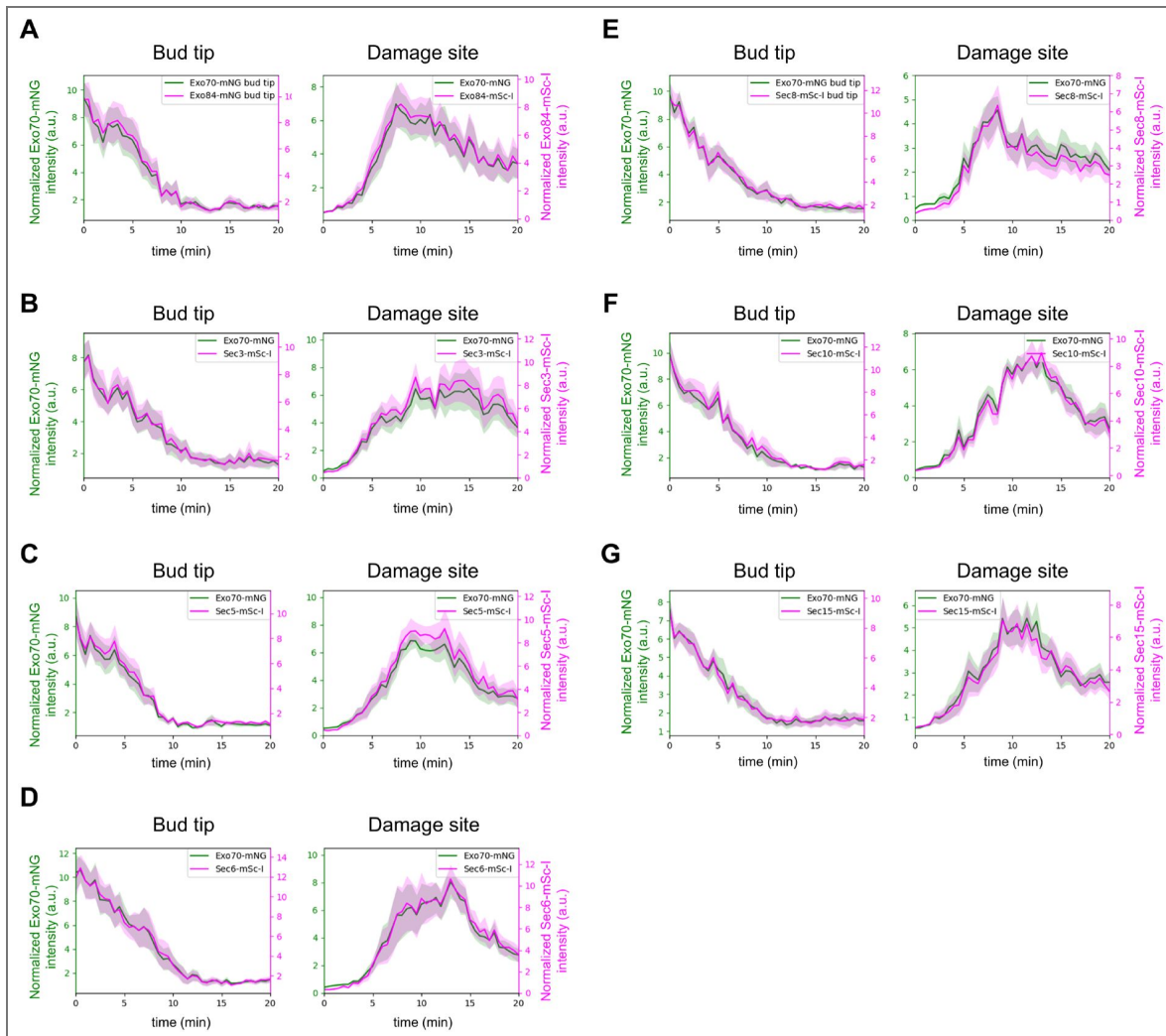


Fig. S3. Accumulation patterns of exocyst subunits.

Fluorescence intensity at the damage site of Exo70-mNG (Green), (A) Exo84-mSc-I, (B) Sec3-mSc-I, (C) Sec5-mSc-I, (D) Sec6-mSc-I, (E) Sec8-mSc-I, (F) Sec10-mSc-I, and (G) Sec15-mSc-I (purple). Lines and shaded regions are the mean and the standard error of the mean. $n = 10$ cells.

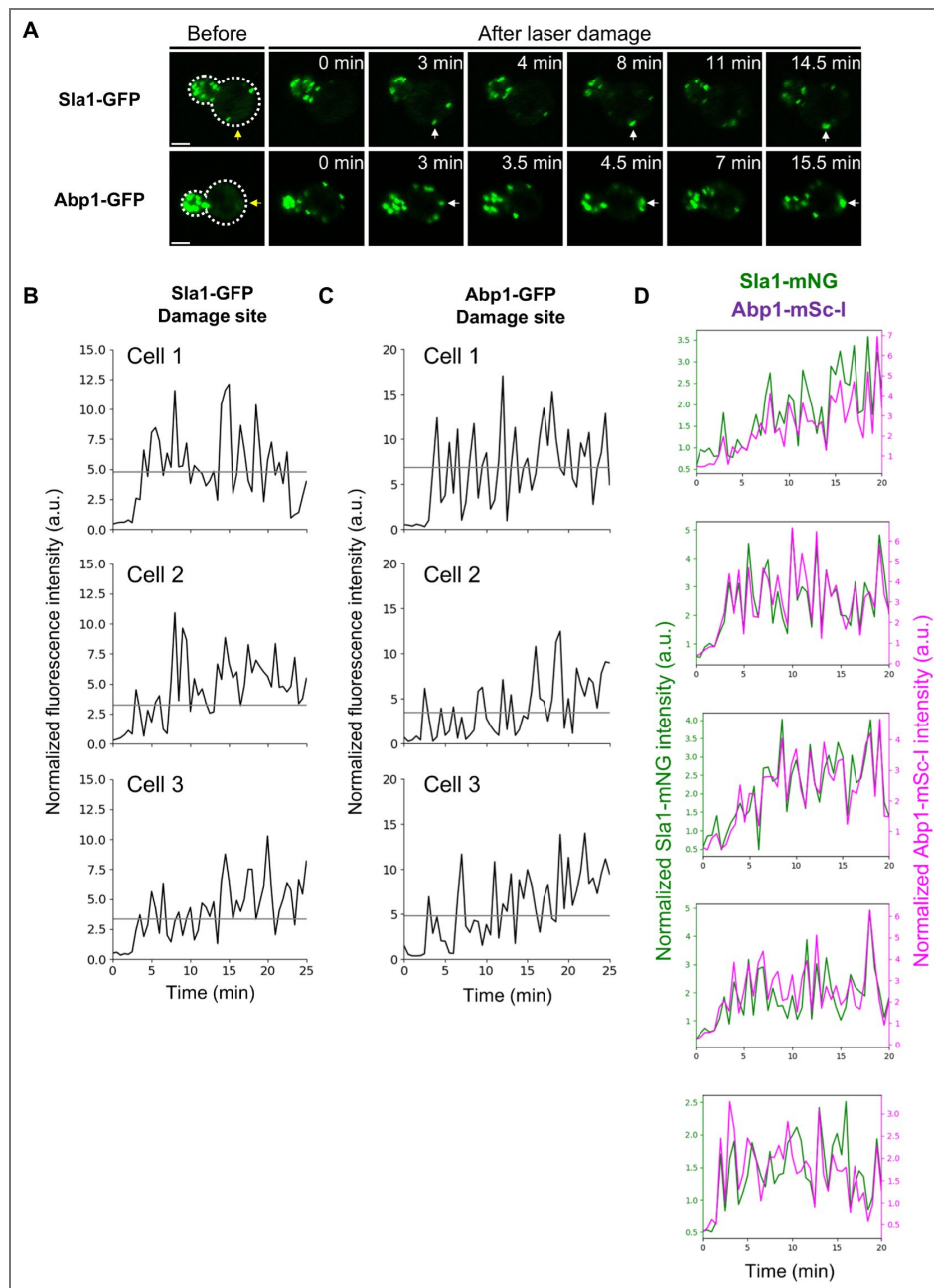


Fig. S4. CME continuously occurs at the damage site.

(A) Representative images of Sla1-GFP and Abp1-GFP. Yellow arrows show the damage site. White arrows showed the recruitment of fluorescence signals. **(B)(C)** Quantification of Sla1-GFP and Abp1-GFP signal at the damage site for three cells. Gray: the fluorescence intensity at the non-damaged site (control) + three times the standard deviation. **(D)** Fluorescence intensity at the damage site of Sla1-mNG (Green) and Abp1-mSc-I (magenta) is shown. $n = 5$ cells.

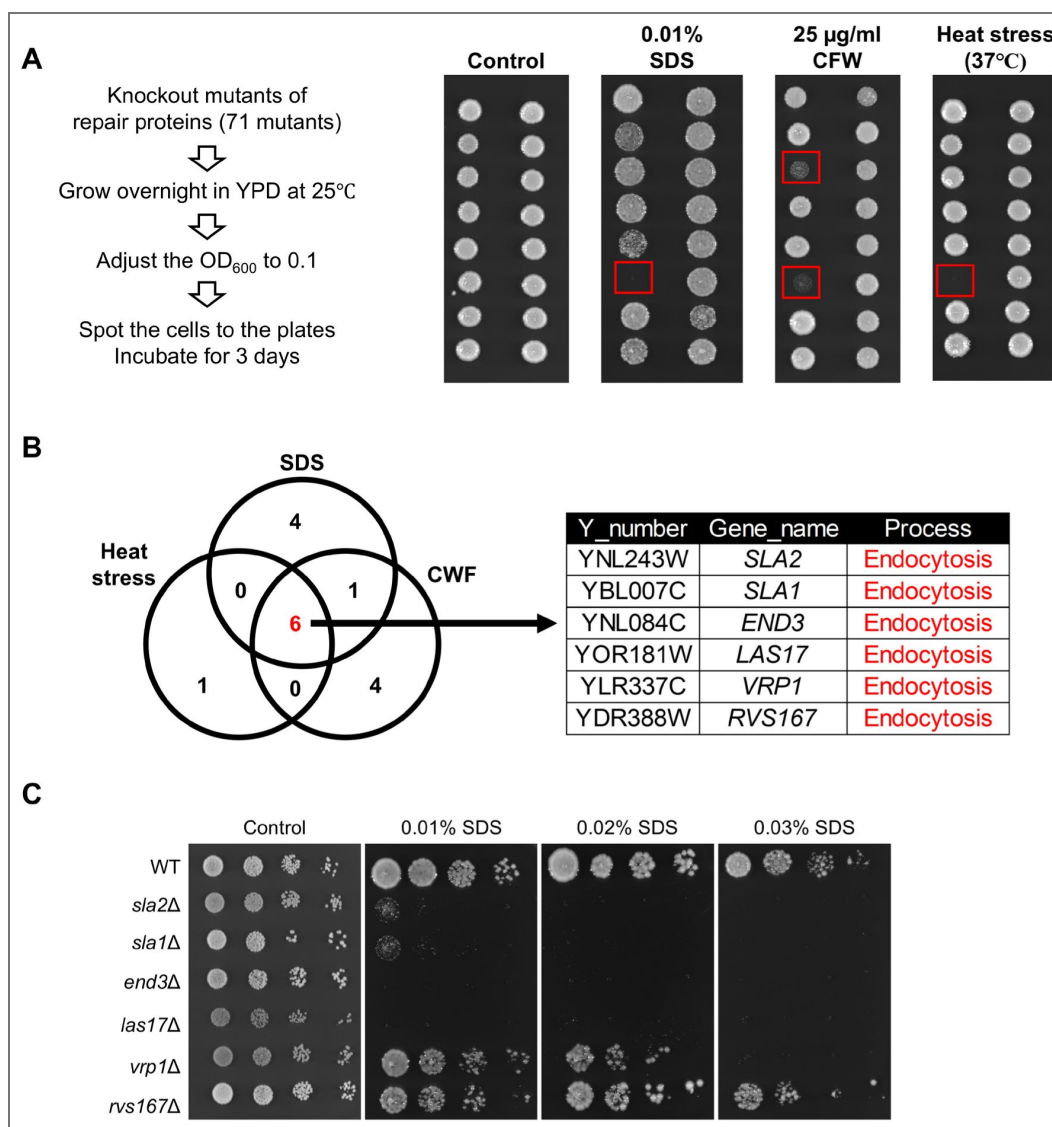


Fig. S5. Growth screening of repair protein knockout mutants in PM/cell wall damage conditions.

(A) Schematic workflow and representative results of growth screening. The red square shows the sensitive mutants. Only the mutants that showed sensitivity to the stresses in two independent colonies were defined as sensitive to the stress. The results of the screening are shown in Data S2. **(B)** The overlap of the screening hits in the three PM/cell wall damage conditions is shown. **(C)** Growth assay of CME mutants. A four-fold serial dilution of the yeast culture was prepared and spotted.

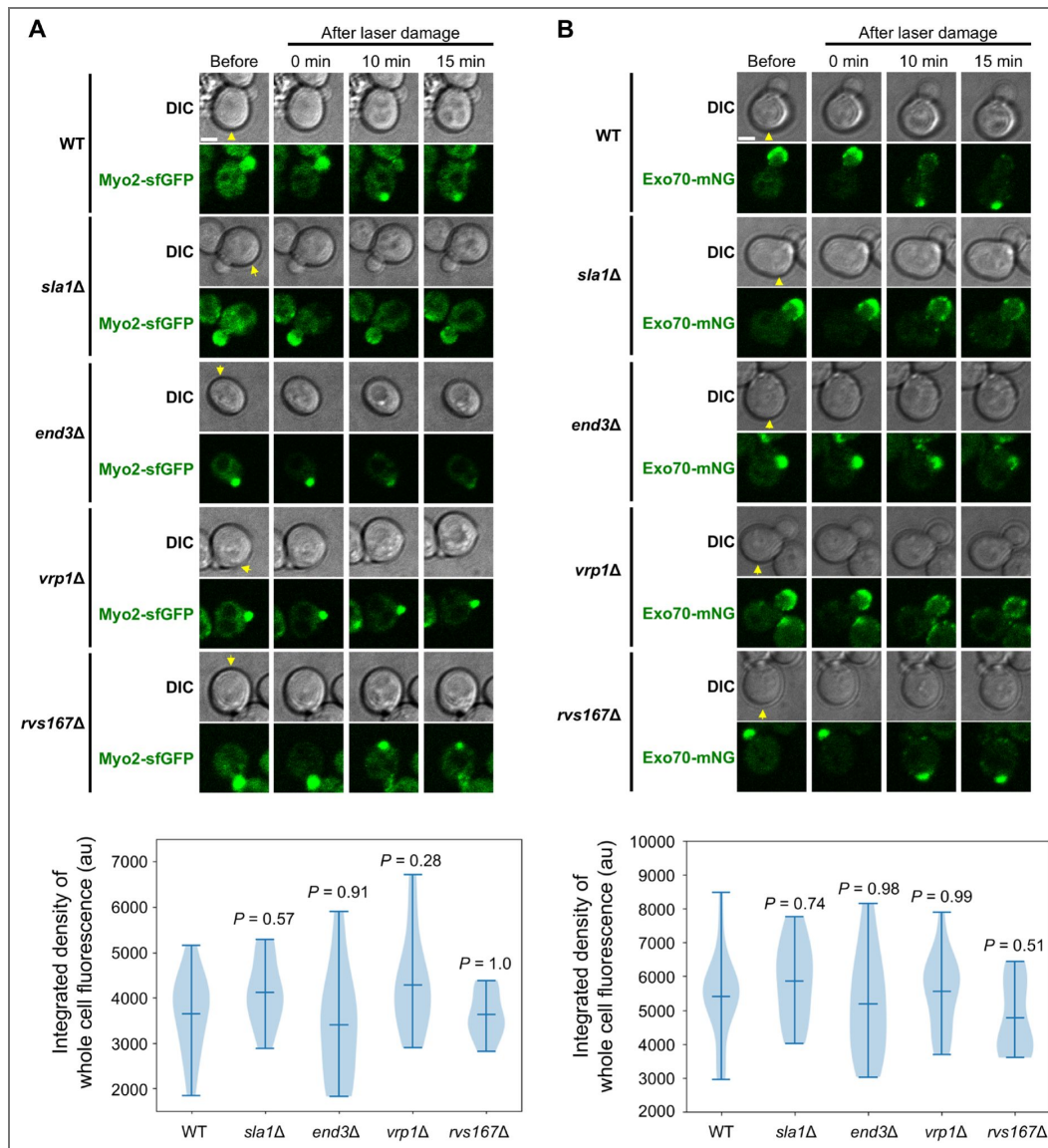


Fig. S6. Representative images of Myo2-sfGFP and Exo70-mNG in CME mutants.

(A) Representative images and whole cell fluorescence quantification of Myo2-sfGFP in the indicated strains. The integrated density of whole cell fluorescence of Myo2-sfGFP before laser damage is compared between WT and CME mutants using Dunnett's test. $n = 13$ for WT, $n = 12$ for *end3Δ*, *sla1Δ*, *rvs167Δ*, and *vrp1Δ*. Scale bar 2 μ m. **(B)** Representative images and whole cell fluorescence quantification of Exo70-mNG in the indicated strains. The integrated density of whole cell fluorescence of Exo70-mNG before laser damage is compared between WT and CME mutants using Dunnett's test. $n = 18$ for WT, $n = 12$ for *end3Δ*, *sla1Δ*, and *rvs167Δ*, $n = 14$ for *vrp1Δ*. Scale bar 2 μ m. Yellow arrows show the damage site.

Fig. S7. CME proteins are not required for the degradation of Bni1-13xMyc and Sec3-GFP after SDS treatment.

(A) Immunoblotting of Bni1-13xMyc before and after SDS treatment (Before (-), 1hr, and 2hr) in WT, *rvs167Δ*, and *end3Δ*. **(B)** Immunoblotting of Sec3-GFP before and after SDS treatment (Before (-), 1hr, and 2hr) in WT, *rvs167Δ*, and *end3Δ*.

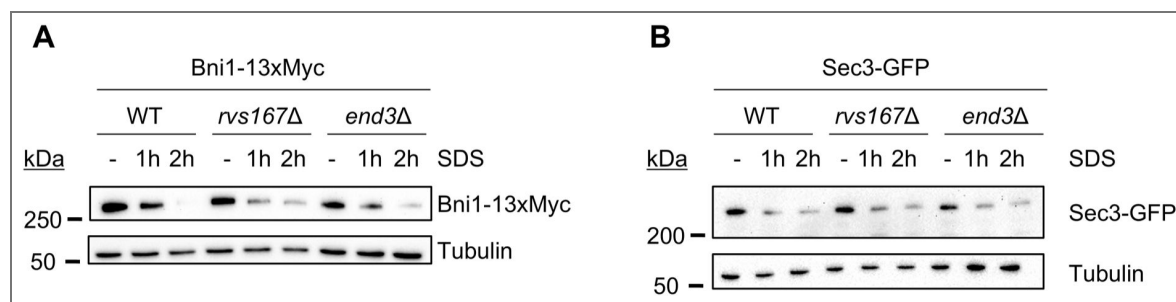


Fig. S8. Fluorescence intensity changes of CME proteins at the bud tip.

(A) Representative images of Apl1-GFP, Ede1-GFP, and Ent1-GFP. **(B)** The ratio of normalized fluorescence intensity at 10 min at the bud tip to the normalized fluorescence intensity at 0 min at the bud tip for all repair protein candidates. Red: CME proteins. The list of CME proteins is adapted from (53). The *P* value was calculated by the two-sided Mann-Whitney U test.

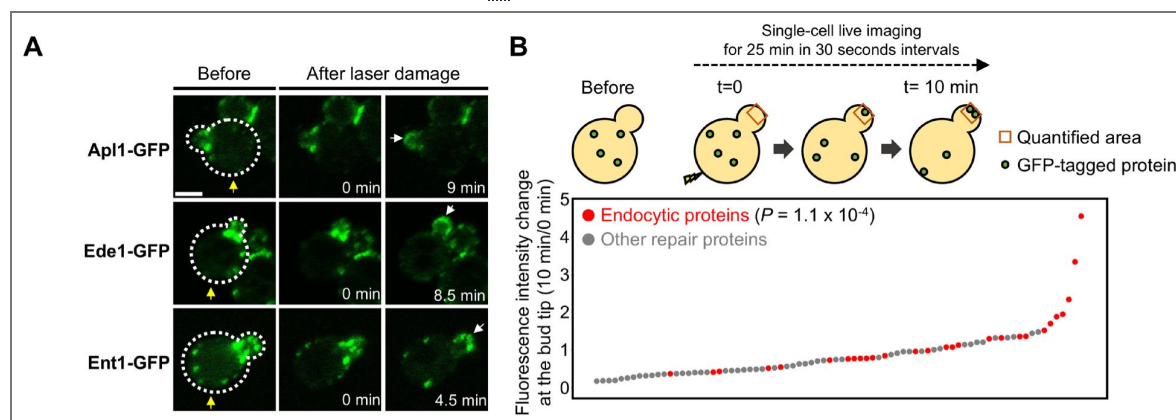
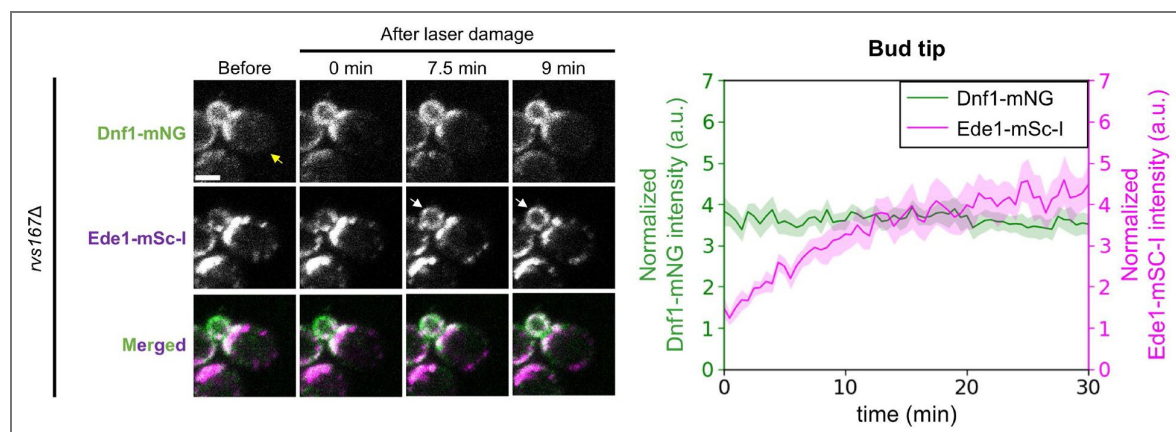


Fig. S9. Time course fluorescence intensity changes of Dnf1-mNG and Ede1-mSc-I at the bud tip in *rvs167Δ*.

Representative images of Dnf1-mNG and Ede1-mSc-I in *rvs167Δ* and normalized fluorescence intensity of Dnf1-mNG (Green) and Ede1-mSc-I (Purple) at the bud tip in *rvs167Δ*. Yellow arrows show the damage site. White arrows show the recruitment of fluorescence signals. Scale bar 2 μ m.



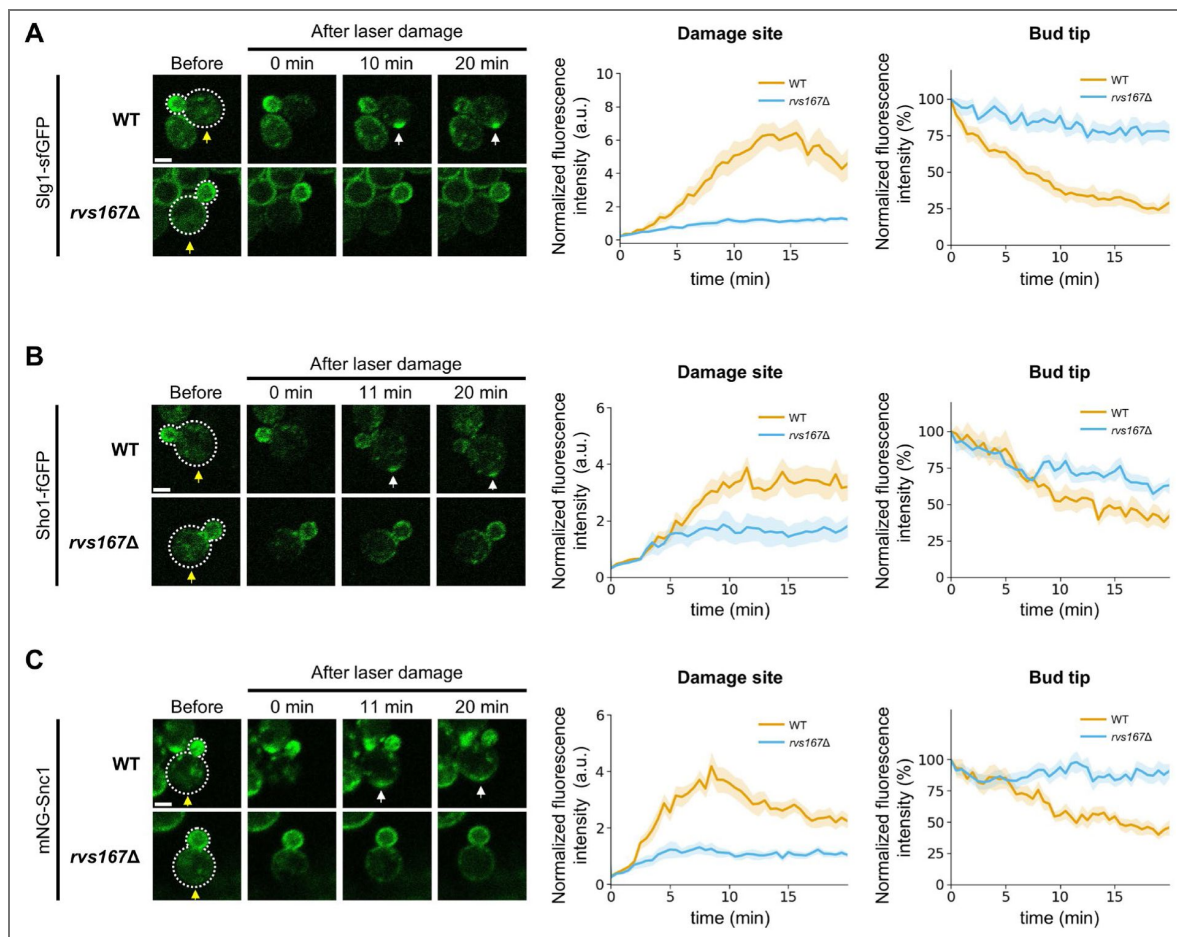


Fig. S10. Time course fluorescence intensity changes of repair proteins with TMDs.

(A) Representative images of Slg1-sfGFP and the normalized fluorescence intensity at the damage site and at the bud tip in WT and *rvs167Δ*. Scale bar 2 μ m. **(B)** Representative images of Sho1-GFP and the normalized fluorescence intensity at the damage site and at the bud tip in WT and *rvs167Δ*. Scale bar 2 μ m. **(C)** Representative images of mNG-Snc1 and the normalized fluorescence intensity at the damage site and at the bud tip in WT and *rvs167Δ*. Scale bar 2 μ m. Yellow arrows show the damage site. White arrows show the recruitment of fluorescence signals. Scale bar 2 μ m.

Data availability

All data needed to evaluate the conclusions in the paper are present in the main text and/or the supplementary materials.

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Author contributions

Conceptualization, Y.Y. and K.K.; Investigation, Y.Y.; Writing – Original Draft, Y.Y.; Writing – Review & Editing, Y.Y. and K.K.; Funding Acquisition, Y.Y. and K.K.; Supervision, K.K.

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Author ORCID iDs

Yuta Yamazaki: <https://orcid.org/0000-0001-5323-6007>

Keiko Kono: <https://orcid.org/0000-0003-4057-8790>

Additional files

[Data_S1-S2_Tables_S1-S3](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

In this manuscript, Yamazaki et al. conducted multiple microscopy-based GFP localization screens, from which they identified proteins that are associated with PM/cell wall damage stress response. Specifically, the authors identified that bud-localized TMD-containing proteins and endocytotic proteins are associated with PM damage stress. The authors further demonstrated that polarized exocytosis and CME are temporally coupled in response to PM damage, and CME is required for polarized exocytosis and the targeting of TMD-containing proteins to the damage site. From these results, the authors proposed a model that CME delivers TMD-containing repair proteins between the bud tip and the damage site.

Strengths:

Overall, this is a well-written manuscript, and the experiments are overall well-conducted. The authors identified many repair proteins and revealed the temporal coordination of different categories of repair proteins. Furthermore, the authors demonstrated that CME is required for targeting of repair proteins to the damage site, as well as cellular survival in response to stress related to PM/cell wall damage. Although the roles of CME and bud-localized proteins in damage repair are not completely new to the field, this work does have conceptual advances by identifying novel repair proteins and proposing the intriguing model that the repairing cargoes are shuttled between the bud tip and the damaged site through coupled exocytosis and endocytosis.

Weaknesses:

While the results presented in this manuscript are convincing, they might not be sufficient to support some of the authors' claims. Especially in the last two result sessions, the authors claimed CME delivers TMD-containing repair proteins from the bud tip to the damage site. The model is no doubt highly possible based on the data, but caveats still exist. For example, the repair proteins might not be transported from one localization to another localization, but are degraded and re-synthesized. Although the Gal-induced expression system can further support the model to some extent, I think more direct verification (such as FLIP or photo-convertible fluorescence tags to distinguish between pre-existing and newly synthesized proteins) would significantly improve the strength of evidence.

Review on revised version:

The authors addressed most of concerns that were originally raised, primarily by revising the text and figures and expanding the discussion, which improves the clarity of the manuscript. Although the authors did not address my major concern on the shuttling/trafficking model experimentally, I do understand the limitation of resources and time. The authors noted that they planned to do these experiments for their future work, and such studies would be more definitive evaluations for the proposed model. Overall I think this is a very interesting and well-conducted study and I enjoyed reading this manuscript. I look forward to their following research of this study.

<https://doi.org/10.7554/eLife.108585.2.sa3>

Reviewer #2 (Public review):

This paper remarkably reveals the identification of plasma membrane repair proteins, revealing spatiotemporal cellular responses to plasma membrane damage. The study highlights a combination of sodium dodecyl sulfate (SDS) and laser for identifying and characterizing proteins involved in plasma membrane (PM) repair in *Saccharomyces cerevisiae*. From 80 PM repair proteins that were identified, 72 of them were novel proteins. The use of both proteomic and microscopy approaches provided a spatiotemporal coordination of exocytosis and clathrin-mediated endocytosis (CME) during repair. Interestingly, the authors were able to demonstrate that exocytosis dominates early and CME later, with CME also playing an essential role in trafficking transmembrane-domain (TMD) containing repair proteins between the bud tip and the damage site.

Weaknesses/limitations:

- Still, there is a lack of clarity about mentioning Pkc1 as the best characterized repair protein, or why is Pkc1 mentioned only as it is changing the localization?!

- The use of a C-terminal GFP-tagged library for the laser damage assay may have limited the identification of proteins whose localization or function depends on an intact N-terminus. N-terminal regions might contain targeting or regulatory elements; therefore, some relevant repair factors may have been missed. Analysis of endogenously N-terminally tagged strains, at least for selected candidates, could help address this limitation.

- The authors appropriately discuss the limitations of SDS- and laser-induced plasma membrane damage, including the possibility that these approaches may not capture proteins involved in other forms of membrane injury, such as mechanical or osmotic stress.

<https://doi.org/10.7554/eLife.108585.2.sa2>

Reviewer #3 (Public review):

Summary:

This work aims to understand how cells repair damage to the plasma membrane (PM). This is important as failure to do so will result in cell lysis and death. Therefore, this is an important fundamental question with broad implications for all eukaryotic cells. Despite this importance, there are relatively few proteins known to contribute to this repair process. This study expands the number of experimentally validated PM from 8 to 80. Further, they use precise laser-induced damage of the PM/cell wall and use live-cell imaging to track the recruitment of repair proteins to these damage sites. They focus on repair proteins that are involved in either exocytosis or clathrin-mediated endocytosis (CME) to understand how these membrane remodeling processes contribute to PM repair. Through these experiments, they find that while exocytosis and CME both occur at the sites of PM damage, exocytosis predominates the early stages of repairs, while CME predominates in the later stages of repairs. Lastly, they propose that CME is responsible for diverting repair proteins localized to the growing bud cell to the site of PM damage.

Strengths:

The manuscript is very well written and the experiments presented flow logically. The use of laser-induced damage and live-cell imaging to validate the proteome-wide screen using SDS induced damage strengthen the role of the identified candidates in PM/cell wall repair.

Comments on revisions:

The authors have very nicely addressed my previous comments and I have no further concerns.

<https://doi.org/10.7554/eLife.108585.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This work provides an important resource identifying 72 proteins as novel candidates for plasma membrane and/or cell wall damage repair in budding yeast, and describes the temporal coordination of exocytosis and endocytosis during the repair process. The data are convincing; however, additional experimental validation will better support the claim that repair proteins shuttle between the bud tip and the damage site.

We thank the editors and reviewers for their positive assessment of our work and the constructive feedback to improve our manuscript. We agree with the assessment that additional validation of repair protein shuttling between the bud tip and the damage site is required to further support the model.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this manuscript, Yamazaki et al. conducted multiple microscopy-based GFP localization screens, from which they identified proteins that are associated with PM/cell wall damage stress response. Specifically, the authors identified that budlocalized TMD-containing proteins and endocytotic proteins are associated with PM damage stress. The authors further demonstrated that polarized exocytosis and CME are temporally coupled in response to PM damage, and CME is required for polarized exocytosis and the targeting of TMD-containing proteins to the damage site. From these results, the authors proposed a model that CME delivers TMD-containing repair proteins between the bud tip and the damage site.

Strengths:

Overall, this is a well-written manuscript, and the experiments are well-conducted. The authors identified many repair proteins and revealed the temporal coordination of different categories of repair proteins. Furthermore, the authors demonstrated that CME is required for targeting of repair proteins to the damage site, as well as cellular survival in response to stress related to PM/cell wall damage. Although the roles of CME and bud-localized proteins in damage repair are not completely new to the field, this work does have conceptual advances by identifying novel repair proteins and proposing the intriguing model that the repairing cargoes are shuttled between the bud tip and the damaged site through coupled exocytosis and endocytosis.

Weaknesses:

While the results presented in this manuscript are convincing, they might not be sufficient to support some of the authors' claims. Especially in the last two result sessions, the authors claimed CME delivers TMD-containing repair proteins from the bud tip to the damage site. The model is no doubt highly possible based on the data, but caveats still exist. For example, the repair proteins might not be transported from one localization to another localization, but are degraded and resynthesized. Although the Gal-induced expression system can further support the model to some extent, I think more direct verification (such as FLIP or photo-convertible fluorescence tags to distinguish between pre-existing and newly synthesized proteins) would significantly improve the strength of evidence.

Major experiment suggestions:

(1) The authors may want to provide more direct evidence for "protein shuttling" and for excluding the possibility that proteins at the bud are degraded and synthesized de novo near the damage site. For example, if the authors could use FLIP to bleach budlocalized fluorescent proteins, and the damaged site does not show fluorescent proteins upon laser damage, this will strongly support the authors' model. Alternatively, the authors could use photo-convertible tags (e.g., Dendra) to differentiate between preexisting repair proteins and newly synthesized proteins.

We thank the reviewer for evaluating our work and giving us important feedback. We agree that the FLIP and photo-convertible experiments will further confirm our model. Here, due to time and resource constraints, we decided not to perform this experiment. Instead, we have discussed this limitation in 363-366. Our proposed model of repair protein shuttling should be further tested in our future work.

(2) In line with point 1, the authors used Gal-inducible expression, which supported their model. However, the author may need to show protein abundance in galactose, glucose, and upon PM damage. Western blot would be ideal to show the level of fulllength proteins, or whole-cell fluorescence quantification can also roughly indicate the protein abundance. Otherwise, we cannot assume that the tagged proteins are only expressed when they are growing in galactose-containing media.

Thank you very much for raising the concern and suggesting the important experiments. We agree that the Western blot experiment to confirm the mNG-Snc1 expression in each medium will further strengthen our conclusion. Along with point (1), further investigation of repair protein shuttling between the bud tip and the damage site and the mechanisms underlying it will be an important future direction. As described above, we have discussed this limitation in 363-366.

(3) Similarly, for Myo2 and Exo70 localization in CME mutants (Figure 4), it might be worth doing a western or whole-cell fluorescence quantification to exclude the caveat that CME deficiency might affect protein abundance or synthesis.

We thank the reviewer for suggesting the point. Following the reviewer's suggestion, we quantified the whole-cell fluorescence of WT and CME mutants and verified that the effect of the CME deletion on the expression levels of Myo2-sfGFP and Exo70-mNG is minimal (Figure S6). We added the description in lines 211-212.

(4) From the authors' model in Figure 7, it looks like the repair proteins contribute to bud growth. Does laser damage to the mother cell prevent bud growth due to the reduction of TMD-containing repair proteins at the bud? If the authors could provide evidence for that, it would further support the model.

Thank you very much for raising the important point. We speculate that the reduction of TMD-containing proteins at the bud by CME is one of the causes of cell growth arrest after PM damage (1). This is because TMD-containing repair proteins at the bud tip, including phospholipid flippases (Dnf1/Dnf2), Snc1, and Dfg5, are involved in polarized cell growth (2-4). This will be an important future direction as well.

(5) Is the PM repair cell-cycle-dependent? For example, would the recruitment of repair proteins to the damage site be impaired when the cells are under alpha-factor arrest?

Thank you for raising this interesting point. Indeed, the senior author Kono previously performed this experiment when she was in David Pellman's lab. The preliminary results suggest that Pkc1 can be targeted to the damage site, without any impairment, under alpha-factor arrest. A more comprehensive analysis in the future will contribute to concluding the relation between PM repair and the cell cycle.

Reviewer #2 (Public review):

*This paper remarkably reveals the identification of plasma membrane repair proteins, revealing spatiotemporal cellular responses to plasma membrane damage. The study highlights a combination of sodium dodecyl sulfate (SDS) and laser for identifying and characterizing proteins involved in plasma membrane (PM) repair in *Saccharomyces cerevisiae*. From 80 PM, repair proteins that were identified, 72 of them were novel*

proteins. The use of both proteomic and microscopy approaches provided a spatiotemporal coordination of exocytosis and clathrin-mediated endocytosis (CME) during repair. Interestingly, the authors were able to demonstrate that exocytosis dominates early and CME later, with CME also playing an essential role in trafficking transmembrane-domain (TMD)containing repair proteins between the bud tip and the damage site.

Weaknesses/limitations:

(1) Why are the authors saying that Pkc1 is the best characterized repair protein? What is the evidence?

We would like to thank the reviewer for taking his/her time to evaluate our work and for valuable suggestions. We described Pkc1 as “best characterized” because it was the first protein reported to accumulate at the laser damage site in budding yeast (5). However, as the reviewer suggested, we do not have enough evidence to describe Pkc1 as “best characterized”. We therefore used “one of the known repair proteins” to mention Pkc1 in the manuscript (lines 90-91).

(2) It is unclear why the authors decided on the C-terminal GFP-tagged library to continue with the laser damage assay, exclusively the C-terminal GFP-tagged library. Potentially, this could have missed N-terminal tag-dependent localizations and functions and may have excluded functionally important repair proteins

Thank you very much for the comments. We decided to use the C-terminal GFP-tagged library for the laser damage assay because we intended to evaluate the proteins of endogenous expression levels. The N-terminal sfGFP-tagged library is expressed by the NOP1 promoter, while the C-terminal GFP-tagged library is expressed by the endogenous promoters. We clarified these points in lines 114-118. We agree with the reviewer on that we may have missed some portion of repair proteins in the N-terminaldependent localization and functions by this approach. Therefore, in our manuscript, we discussed these limitations in lines 281-289.

(3) The use of SDS and laser damage may bias toward proteins responsive to these specific stresses, potentially missing proteins involved in other forms of plasma membrane injuries, such as mechanical, osmotic, etc.). SDS stress is known to indirectly induce oxidative stress and heat-shock responses.

Thank you very much for raising this point. We agree that the combination of SDS and laser may be biased to identify PM repair proteins. Therefore, in the manuscript, we discussed this point as a limitation of this work in lines 292-298.

(4) It is unclear what the scale bars of Figures 3, 5, and 6 are. These should be included in the figure legend.

We apologize for the missing scale bars. We added them to the legends of the figures in the manuscript.

(5) Figure 4 should be organized to compare WT vs. mutant, which would emphasize the magnitude of impairment.

Thank you for raising this point. Following the suggestion, we updated Figure 4. In the Figure 4, we compared WT vs mutant in the manuscript. We clarified it in the legends in the manuscript.

(6) It would be interesting to expand on possible mechanisms for CME-mediated sorting and retargeting of TMD proteins, including a speculative model.

Thank you very much for this important suggestion. We think it will be very important to characterize the mechanism of CME-mediated TMD protein trafficking between the bud tip and the damage site. In the manuscript, we discussed the possible mechanism for CME activation at the damage site in lines 328-333. We speculate that the activation of the CME may facilitate the retargeting of the TMD proteins from the damage site to the bud tip.

We do not have a model of how CMEs activate at the bud tip to sort and target the TMD proteins to the damage site. One possibility is that the cell cycle arrest after PM damage (1) may affect the localization of CME proteins because the cell cycle affects the localization of some of the CME proteins (6). We will work on the mechanism of repair protein sorting from the bud tip to the damage site in our future work.

Reviewer #3 (Public review):

Summary:

This work aims to understand how cells repair damage to the plasma membrane (PM). This is important, as failure to do so will result in cell lysis and death. Therefore, this is an important fundamental question with broad implications for all eukaryotic cells. Despite this importance, there are relatively few proteins known to contribute to this repair process. This study expands the number of experimentally validated PM from 8 to 80. Further, they use precise laser-induced damage of the PM/cell wall and use live-cell imaging to track the recruitment of repair proteins to these damage sites. They focus on repair proteins that are involved in either exocytosis or clathrin-mediated endocytosis (CME) to understand how these membrane remodeling processes contribute to PM repair. Through these experiments, they find that while exocytosis and CME both occur at the sites of PM damage, exocytosis predominates in the early stages of repairs, while CME predominates in the later stages of repairs. Lastly, they propose that CME is responsible for diverting repair proteins localized to the growing bud cell to the site of PM damage.

Strengths:

The manuscript is very well written, and the experiments presented flow logically. The use of laser-induced damage and live-cell imaging to validate the proteome-wide screen using SDS-induced damage strengthens the role of the identified candidates in PM/cell wall repair.

Weaknesses:

(1) Could the authors estimate the fraction of their candidates that are associated with cell wall repair versus plasma membrane repair? Understanding how many of these proteins may be associated with the repair of the cell wall or PM may be useful for thinking about how these results are relevant to systems that do or do not have a cell wall. Perhaps this is already in their GO analysis, but I don't see it mentioned in the manuscript.

We would like to thank the reviewer for taking his/her time to evaluate our work and valuable suggestions. We agree that this is important information to include. Although it may be difficult to completely distinguish the PM repair and cell wall repair proteins, we have identified at least six proteins involved in cell wall synthesis (Flc1, Dfg5, Smi1, Skg1, Tos7, and Chs3). We included this information in lines 142-146 in the manuscript.

(2) Do the authors identify actin cable-associated proteins or formin regulators associated with sites of PM damage? Prior work from the senior author (reference 26) shows that the formin Bnr1 relocalizes to sites of PM damage, so it would be interesting

if Bnr1 and its regulators (e.g., Bud14, Smy1, etc) are recruited to these sites as well. These may play a role in directing PM repair proteins (see more below).

Thank you for the suggestion. We identified several Bnr1-interacting proteins, including Bud6, Bil1, and Smy1 (Table S2), although Bnr1 itself was not identified in our screening. This could be attributed to the fact that (1) C-terminal GFP fusion impaired the function of Bnr1, and (2) a single GFP fusion is not sufficient to visualize the weak signal at the damage site. Indeed, in reference 26, 3GFP-Bnr1 (N-terminal 3xGFP fusion) was used.

(3) Do the authors suspect that actin cables play a role in the relocalization of material from the bud tip to PM damage sites? They mention that TMD proteins are secretory vesicle cargo (lines 134-143) and that Myo2 localizes to damage sites. Together, this suggests a possible role for cable-based transport of repair proteins. While this may be the focus of future work, some additional discussion of the role of cables would strengthen their proposed mechanism (steps 3 and 4 in Figure 7).

Thank you very much for the suggestion. We agree that actin cables may play a role in the targeting of vesicles and repair proteins to the damage site. Following the reviewer's suggestion, we discussed the roles of Bnr1 and actin cables for repair protein trafficking in lines 309-313 in the manuscript.

(4) Lines 248-249: I find the rationale for using an inducible Gal promoter here unclear. Some clarification is needed.

Thank you for raising this point. We clarified this as possible as we could in lines 249-255 in the manuscript.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) The N-terminal GFP collection screen is interesting but seems irrelevant to the rest of the results. The authors discussed that in the discussion part, but it might be worth showing how many hits from the laser damage screen (in Figure 2) overlap with the Nterminal GFP screen hits.

Thank you for the suggestion. We found that 48 out of 80 repair proteins are hits in the N-terminal GFP library (Table S1 and S2). This result suggested that the N-terminal library is also a useful resource for identifying repair proteins. In the manuscript, we discussed it in lines 288-289.

(2) SDS treatment seems a harsh stressor. As the authors mentioned, the overlapped hits from the N- and C-terminal GFP screen might be more general stress factors. Thus, I think Line 84 (the subtitle) might be overclaiming, and the authors might need to tone down the sentence.

Thank you for the suggestion. Following the reviewer's suggestion, we changed the sentence to "Proteome-scale identification of SDS-responsive proteins" in the manuscript. We believe that the new sentence describes our findings more precisely.

(3) Line 103-106, it does not seem obvious to me that the protein puncta in the cytoplasm are due to endocytosis. The authors might need to provide more experimental evidence for the conclusion, or at least provide more reasoning/references on that aspect (e.g., several specific protein hits belonging to that group have been shown to be endocytosed).

Thank you very much for raising this point. We agree with the reviewer and deleted the description that these puncta are due to endocytosis in the manuscript.

(4) For Figure 1D and S1C, the authors annotated some of the localization changes clearly, but some are confusing to me. For example, "from bud tip/neck" to where? And from where to "Puncta/foci"? A clearer annotation might help the readers to understand the categorization.

Thank you very much for the suggestion. These annotations were defined because it is difficult to conclusively describe the protein localization after SDS treatment. To convincingly identify the destination of the GFP fusion proteins, the dual color imaging of proteins with organelle markers or deep learning-based localization estimation is required. We feel that this might be out of the scope of this work. Therefore, as criteria, we used the localization of protein localization in normal/non-stressed conditions reported in (7) and the Saccharomyces Genome Database (SGD). We clarified this annotation definition in the manuscript (lines 413-436).

(5) For localization in Figure 2C, as I understand, does it refer to the "before damage/normal" localization? If so, I think it would be helpful to state that these localizations are based on the untreated/normal conditions in the text.

Yes, it refers to the “before damage/normal localization”. Following the reviewer’s suggestion, we stated that these localizations are based on these conditions in the manuscript (line 130).

(6) The authors mentioned "four classes" in Line 120, but did not mention the "PM to cytoplasm" class in the text. It would be helpful to discuss/speculate why these transporters might contribute to PM damage repair.

Thank you very much for this suggestion. We speculated that these transporters are endocytosed after PM damage because endocytosis of PM proteins contributes to cell adaptation to environmental stress (8). We mentioned it in the manuscript (lines 120-122).

(7) Line 175-180 My understanding of the text is that the signals of Exo70-mNG/Dnf1mNG peak before the Ede1-mSc-I peaks. They occur simultaneously, but their dominating phase are different. It is clearer when looking at the data, but I think the conclusion sentences themselves are confusing to me. The authors might consider rewriting the sentences to make them more straightforward.

Thank you very much for pointing this out. Following the reviewer’s suggestion, we revised the sentence (lines 177-182 in the manuscript).

Reviewer #2 (Recommendations for the authors):

It would be interesting to expand on the functional characterization of the 72 novel candidates and explore possible mechanisms for CME-mediated sorting and retargeting of TMD proteins by including a speculative model.

Thank you very much for the comment. We agree that the further characterization of novel repair proteins and exploration of the possible mechanisms for CME-mediated TMD protein sorting and retargeting are truly important. This should be our important future direction.

Reviewer #3 (Recommendations for the authors):

The x-axis in Figure 1C is labeled 'Ratio' - what is this a ratio of?

Thank you for raising this point. It is the ratio of the number of proteins associated with a GO term to the total number of proteins in the background. We clarified it in the legend of Figure 1C in the manuscript.

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